Array 28 (2025) 100529

Contents lists available at ScienceDirect

Array

journal homepage: www.elsevier.com/locate/array

Crocodile Ambush Optimization Algorithm: A new bio-inspired metaheuristic algorithm for solving optimization problems

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ARTICLE INFO

Keywords:
Optimization
Metaheuristics
Algorithm
Bio-inspired

ABSTRACT

This paper presents the Crocodile Ambush Optimization Algorithm (CAOA), a novel metaheuristic inspired by the energy-saving and ambush hunting behavior of crocodiles. CAOA integrates adaptive energy decay modeling, stochastic leader selection, and threshold-based solution reinitialization to achieve a balanced trade-off between exploration and exploitation. The algorithm is evaluated on a subset of 29 representative benchmark functions selected from the CEC 2017 test suite under various dimensional settings (30, 50, 100), and its performance is compared against several classical and modern algorithms, including PSO, GWO, DE, SCA, WOA, SSA, HHO, and HGS. Experimental results show that CAOA achieves superior or competitive convergence accuracy across unimodal, multimodal, and composite functions, while maintaining a lower average runtime. Furthermore, CAOA is applied to two real-world constrained engineering design problems, where it successfully identifies feasible, high-quality solutions under nonlinear constraints. These results demonstrate that CAOA is a robust, efficient, and adaptable optimizer for both benchmark and real-world problems.

1. Introduction

An optimization problem is one with more than one solution that lies in the feasible region, and the selection of the best solution among them is what is known as optimization [1]. An optimization problem typically consists of three main components: decision variables [2], constraints [3], and an objective function [4]. As science and technology continue to develop, optimization problems grow more complex, and new optimization challenges arise, requiring proper tools for solving them.

Optimization methods are generally classified into the following two main categories: deterministic methods [5] and stochastic methods [6]. Deterministic methods can be classified further into gradient-based and non-gradient-based methods, and generally, they perform well for linear, convex, and somewhat simple optimization problems. They often lose efficiency while facing complex, nonlinear problems that are- non-differentiable, non-convex, and NP-hard [7] — characteristics frequently arising in real-world applications. Such restrictions upon deterministic methods propelled researchers within projects focused on stochastic approaches, specifically metaheuristic algorithms [8].

Metaheuristic algorithms are potent and promising in solving a wide range of optimization problems through random operators, trial-and-error processes, and random exploration of the search space. Their simplicity and ease of implementation, independence from derivative information, and tendency to work efficiently in high dimensionality and non-linear situations have increased their popularity and application [9].

The optimization process in metaheuristic algorithms typically begins by initializing a set of randomly generated feasible solutions within the search space. These candidate solutions are iteratively updated and improved based on the algorithm's specific rules. After completing a predefined number of iterations, the best candidate solution is selected as the final solution to the problem. However, it is important to note that the solutions obtained by metaheuristic algorithms are not guaranteed to be the global optimum. Due to their inherent randomness, the solutions are considered quasi-optimal [10] — often close to, and sometimes equal to, the global optimum. The differences in the performance of metaheuristic algorithms when applied to similar problems can be attributed to the variations in their search and update mechanisms. This ongoing challenge, along with the desire to achieve better solutions, has driven the continuous design and development of novel optimization algorithms.

A crucially researched question regarding metaheuristic algorithms is whether the birth of new optimization methods is worth it, given the existing number of algorithms. A general answer should come from the No Free Lunch theorem [11], which explains that better performance

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https://doi.org/10.1016/j.array.2025.100529
Received 2 April 2025; Received in revised form 26 September 2025; Accepted 29 September 2025
Available online 10 October 2025
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of an algorithm over a certain optimization set does not lead to the same level of performance over other kinds of problem instances. Hence no particular algorithm could always outdo others in all optimization tasks, and continues to give credence even now to the search for newer algorithms.

In relation to this motivation, the paper presents and devises a new optimization algorithm, called the Crocodile Ambush Optimization Algorithm (CAOA), which mimics the natural behavior of coatis. The main contributions of this work can be summarized as follows:

- A novel bio-inspired metaheuristic algorithm, Crocodile Ambush Optimization Algorithm (CAOA), is proposed, inspired by the energy-aware ambush hunting strategy of crocodiles, which incorporates adaptive energy decay, stochastic leader selection, and ambush-based position updates.
- A detailed mathematical formulation and pseudocode are presented to describe the dynamic search behavior of CAOA, enabling effective balance between exploration and exploitation.
- The proposed algorithm is benchmarked against thirteen standard functions from the 2017 Congress on Evolutionary Computation (CEC2017) suite, showing superior or competitive performance compared to recent bio-inspired algorithms including AVOA, GTO, HLOA, MGO, FFA, and POA.
- The CAOA is also applied to two constrained real-world engineering problems, demonstrating its effectiveness in solving practical optimization tasks with multiple design constraints.

The proposed CAOA has some striking advantages when it comes to global optimization problems. First of all, CAOA was simple in design with very few control parameters, utterly avoiding cumbersome parameter tuning. Second, it could bring forth high efficiency when solving optimization problems in varying fields, including higher dimensions and nonlinear problems. Third, with CAOA, balance would be achieving between exploration and exploitation in the search process to ensure convergence to good-quality solutions is fast. Last but not least, robust and versatile performance in resolving real-world scenarios in an optimization problem makes it an equally attractive alternative.

The rest of this paper is organized as follows: Section 2 presents the literature review. Section 3 introduces the proposed CAOA and its underlying mathematical model. The experimental setup, results and analysis are provided in Section 4. Section 5 analyze CAOA's performance in practical optimization problems. Finally, Section 6 concludes the paper and suggests directions for future research.

2. Literature review

2.1. Overview of deterministic optimization algorithms

They are simply defined as a collection of rules or equations that determine the procedure of solving a particular problem, in an iterative manner. For the same initial conditions, the same output is produced every time the algorithm is run. Randomness is not incorporated into the attack procedure. The structures are deterministically popular in applications that require precision and the utmost reliability.

Classical deterministic optimization methods include gradient-based techniques [12] such as Gradient Descent (GD) [13], Newton's Method [14], and Quasi-Newton Methods [15] like the Broyden–Fletcher–Goldfarb–Shanno (BFGS) [16] algorithm. These methods rely on the gradient [17] or Hessian information [18] of the objective function to guide the search direction and step size. While effective for convex or well-behaved functions, their performance degrades when dealing with non-convex [19], discontinuous [20], or noisy objective functions [21].

Other notable deterministic methods include Linear Programming (LP) [22] and Quadratic Programming (QP) [23], which solve optimization problems defined by linear or quadratic objective functions and constraints, often using algorithms like the Simplex Method [24] or Interior-Point Methods [25]. For combinatorial problems, Branch and Bound (B&B) [26] systematically explore the solution space, guaranteeing global optimality given sufficient time and computational resources.

However, the very basis of the precise nature of deterministic algorithms is that they easily fall victim to local optimums and that their optimization problem functions are usually smooth, continuous, and differentiable. This weakness is now receiving increasing attention with stochastic and hybrid methods having facilitated more flexible approaches to solving intractable high-dimensional problems.

2.2. Overview of stochastic optimization algorithms

Stochastic optimization algorithms are widely used to tackle complex optimization problems due to their ability to escape local optima and explore high-dimensional search spaces. These algorithms can be broadly classified into two main categories: individual-based [27] and population-based algorithms [28].

Individual-based algorithms operate by iteratively improving a single candidate solution. Algorithms such as Simulated Annealing (SA) [29] and Tabu Search (TS) [30] fall into this category. These methods generally have low computational costs but are prone to premature convergence due to their limited exploration capabilities.

On the other hand, population-based algorithms work with a set of candidate solutions and leverage information exchange between them to guide the search process. This category includes a variety of algorithms inspired by natural and physical phenomena.

Evolutionary-based algorithms [31], such as Genetic Algorithm (GA) [32], Differential Evolution (DE) [33], Biogeography-Based Optimization (BBO) [34], Evolution Strategy (ES) [35], and Memetic Algorithm (MA) [36], mimic natural selection processes to evolve solutions over generations.

Swarm intelligence-based algorithms [37], including Grey Wolf Optimizer (GWO) [38], Ant Colony Optimization (ACO) [39], Firefly Algorithm (FA) [40], Bat Algorithm (BA) [41], and Monkey King Evolution Algorithm (MKEA) [42], are inspired by the collective behavior of social creatures.

Additionally, physics-based algorithms [43] like Gravitational Search Algorithm (GSA) [44], Charged System Search (CSS) [45], Electromagnetic Field Optimization (EFO) [46], Thermal Exchange Optimization (TEO) [47], Black Hole Algorithm (BHA) [48], and Galaxy-based Search Algorithm (GbSA) [49] draw analogies from physical laws and processes.

More recently, hybrid algorithms such as Hybrid Particle Swarm Optimization (HPSO) [50] have emerged, combining strengths from multiple paradigms to enhance search efficiency.

Beyond these mainstream categories, a number of novel metaheuristic frameworks have been proposed, drawing inspiration from a wider spectrum of natural and cultural processes. These include:

- **Plant-based algorithms**, such as the Root Growth Algorithm (RGA) [51], which emulate the foraging patterns of plant roots in search of nutrients.
- **Chemistry-based algorithms**, such as Chemical Reaction Optimization (CRO) [52], which simulate molecular interactions and reaction kinetics.
- **Music-inspired algorithms**, such as Harmony Search (HS) [53], based on the improvisation process of musicians to iteratively improve solution quality.
- **Sport-based algorithms**, such as the Football Game Algorithm (FGA) [54], which model strategic decision-making and adaptive role switching in competitive environments.

These expanding categories highlight the continuous innovation in metaheuristic design and reinforce the relevance of exploring new behavioral models — such as the ambush tactics of crocodiles — toward developing efficient optimization algorithms.

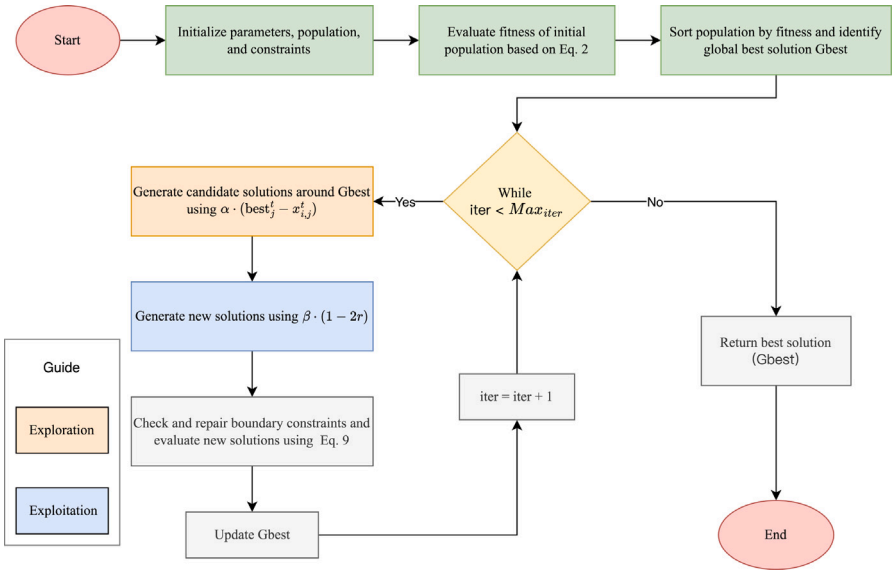

Fig. 1. The framework of Crocodile Ambush Optimization Algorithm.

2.3. Recent nature-inspired metaheuristics

In recent years, a surge of bio-inspired metaheuristics has been proposed to enhance the capabilities of classical algorithms in solving complex, high-dimensional optimization problems. These methods often combine unique biological metaphors with adaptive mechanisms to achieve better exploration–exploitation balance.

Examples include the African Vultures Optimization Algorithm (AVOA) [55], which mimics the scavenging behavior of vultures with a dynamic leader–follower mechanism; the Artificial Gorilla Troops Optimizer (GTO) [56], inspired by gorilla social hierarchy and group-based movement; the Mountain Gazelle Optimizer (MGO) [57], based on the escape and foraging behaviors of gazelles; and the Farmland Fertility Algorithm (FFA) [58], which models soil improvement and fertility dynamics. Additionally, the Puma Optimizer (POA) [59] and Horned Lizard Optimization Algorithm (HLOA) [60] offer novel search strategies rooted in predatory efficiency and adaptive behavior.

These algorithms have shown promising results in engineering design, data clustering, feature selection, and energy-aware scheduling. However, many of them rely heavily on static parameters or lack detailed modeling of dynamic interactions between agents, which limits their flexibility and adaptability in highly constrained or stochastic environments. This motivates the development of the Crocodile Ambush Optimization Algorithm (CAOA), which explicitly integrates dynamic energy management, stochastic leader selection, and adaptive ambush positioning.

3. Crocodile Ambush Optimization Algorithm (CAOA)

The Crocodile Ambush Optimization Algorithm (CAOA) is a novel metaheuristic optimization approach inspired by the ambush hunting strategy of crocodiles. By incorporating low-energy waiting, high-reward hunting, and adaptive behaviors into its design, CAOA effectively balances exploration and exploitation to solve complex optimization problems. The framework of the Crocodile Ambush Optimization Algorithm is shown in Fig. 1.

3.1. Algorithm description and pseudocode

The Crocodile Ambush Optimization Algorithm (CAOA) is a population-based metaheuristic inspired by the predatory behavior of crocodiles during ambush hunting. The algorithm simulates how crocodiles dynamically conserve energy, select ambush points, and launch attacks on prey based on environmental stimuli.

Initially, a population of N individuals is randomly generated within the problem's search space. Each individual represents a potential solution, and is assigned an initial energy value. The fitness of each individual is evaluated using the objective function.

In each iteration, the algorithm selects a leader individual based on an inverse fitness-based probability model. This leader guides the population by influencing the direction and intensity of exploration. Other individuals update their positions based on a balance of attraction to the leader and stochastic perturbation, controlled by parameters α and β.

To simulate energy dynamics, the energy level of each crocodile is decreased proportionally to its movement distance. When an individual's energy drops below a threshold, it is reinitialized, mimicking a crocodile repositioning after depletion. Additionally, if an individual's fitness deteriorates beyond a threshold δ, it may be reinitialized to encourage diversity and avoid local optima.

Throughout the optimization process, the best solution found so far is tracked and updated. A convergence curve is recorded to visualize the optimization progress.

This mechanism ensures a balance between exploitation (focused search near the leader) and exploration (through energy depletion and stochastic behaviors), making CAOA robust for complex and high-dimensional optimization problems.

3.2. Problem description

In many real-world scenarios, optimization problems can be represented as minimizing an objective function $f(x)$, which evaluates the quality of a solution x. In CAOA, crocodiles symbolize candidate solutions, and their goal is to find the optimal ambush point corresponding to the best solution. Mathematically, this is formulated as:

$$\min_{x \in S} f(x), \tag{1}$$

where S is the search space. For example, in a scheduling problem, x may represent the sequence of tasks, and $f(x)$ quantifies the total completion time. The solution x^* that minimizes $f(x)$ represents the optimal configuration, such that $f(x^*) \leq f(x)$ for all $x \in S$.

To solve this problem efficiently, CAOA employs a population-based approach, with each crocodile exploring S in parallel. This strategy mirrors the behavior of crocodiles in nature, where they spread out across potential hunting grounds to maximize their collective success.

Algorithm 1: Crocodile Ambush Optimization Algorithm (CAOA)

Input: Population size N, max iterations T, lower bound lb, upper bound ub, dimension D, objective function $f(\cdot)$, parameters α, β, γ, δ, initial energy E_0
Output: Global best solution $gBest$ and its score $gBestScore$
Initialize population positions $x_i \in [lb, ub]^D$, $\forall i = 1, \ldots, N$;
Set initial energies $E_i = E_0$, compute initial fitness $f(x_i)$;
Set global best $gBest$ and $gBestScore$;
for $t = 1$ **to** T **do**
  Store previous positions and fitness values;
  Select leader based on inverse fitness probability;
  for $i = 1$ **to** N **do**
    if *i is leader* **then**
      ;
    Generate random vector r;
    Update position: $x_i \leftarrow x_i + \alpha(x_{leader} - x_i) + \beta(1 - 2r)$;
    Apply boundary control to x_i;
    Compute new fitness $f_{new} = f(x_i)$;
    if $|f_{new} - f_{old}| > \delta$ ***and*** $f_{new} > f_{old}$ **then**
      Reinitialize x_i randomly within bounds;
      Recompute f_{new};
    Update x_i and $f(x_i)$;
  Compute energy decay: $E_i \leftarrow E_i - \gamma \cdot \text{distance}(x_i, x_i^{old})$;
  if $E_i \leq 0$ **then**
    Reinitialize x_i and E_i; Recompute fitness;
  Update global best if better solution found;
  Record $gBestScore$ in convergence curve;
return $gBest$, $gBestScore$, convergence history;

3.3. Representation of solutions

Each crocodile in CAOA represents a candidate solution x within the search space S. The structure of x depends on the problem domain. For instance: In a path-planning problem, x is a sequence of coordinates representing a path. In a resource allocation problem, x represents the distribution of resources. In high-dimensional optimization, $x = (x_1, x_2, \ldots, x_n)$ is a continuous vector of decision variables.

This flexibility allows CAOA to be applied across diverse problem domains. The algorithm maintains a population $P = \{x_1, x_2, \ldots, x_N\}$, where N is the number of crocodiles, and iteratively updates their positions to improve the objective function $f(x)$.

3.4. Initialization of the population

The algorithm begins by initializing the crocodiles' positions randomly within the bounds of the search space. The position $x_{i,j}$ of the ith crocodile in the jth dimension is given by:

$$x_{i,j} = \text{lb}_j + r \cdot (\text{ub}_j - \text{lb}_j), \tag{2}$$

where lb_j and ub_j are the lower and upper bounds of the jth dimension, and r is a random number uniformly distributed in $[0, 1]$. This diverse placement ensures that the crocodiles explore the search space effectively from the start.

The initial objective values $f(x_i)$ are computed for all crocodiles, and the one with the lowest $f(x)$ is selected as the leader. The leader's position serves as a reference for guiding the population during subsequent iterations.

3.5. Energy dynamics

In nature, crocodiles are known for their ability to conserve energy while waiting at ambush points. This conservation strategy allows them to minimize unnecessary energy expenditure until an opportunity to attack arises. However, their energy levels naturally decrease during hunting and movement. To model this behavior in CAOA, each crocodile is assigned an energy level E_i, which determines its capacity for exploration and exploitation in the search space. This energy level is updated dynamically at each iteration to reflect the costs of movement.

Initially, all crocodiles are assigned the same energy level to ensure a uniform starting condition for the population. Mathematically, this initialization is expressed as:

$$E_i = E_{\text{init}}, \quad \forall i \in \{1, 2, \ldots, N\}. \tag{3}$$

Here, E_{init} is a predefined constant representing the initial energy level of each crocodile, and N is the total number of crocodiles in the population. This uniform initialization ensures that every crocodile starts with equal potential for exploration, allowing the algorithm to distribute the search effort evenly across the solution space.

As the algorithm progresses, crocodiles move within the search space to explore and exploit regions of interest. This movement incurs an energy cost, which is proportional to the distance traveled. The energy update rule is given by:

$$E_i^{t+1} = E_i^t - \gamma \cdot \|\Delta x_i\|, \tag{4}$$

where E_i^{t+1} and E_i^t represent the energy levels of the ith crocodile at iterations $t+1$ and t, respectively. The term γ is an energy consumption coefficient that governs how much energy is depleted per unit distance traveled, while $\|\Delta x_i\|$ quantifies the Euclidean distance the crocodile has moved during the current iteration.

The Euclidean distance $\|\Delta x_i\|$ is calculated as:

$$\|\Delta x_i\| = \sqrt{\sum_{j=1}^{n} (x_{i,j}^{t+1} - x_{i,j}^t)^2}. \tag{5}$$

In this equation, n is the dimensionality of the search space, and $x_{i,j}^{t+1}$ and $x_{i,j}^t$ denote the positions of the ith crocodile in the jth dimension at iterations $t+1$ and t, respectively. The Euclidean distance captures the overall movement of the crocodile across all dimensions, ensuring that longer movements result in higher energy consumption.

The energy dynamics play a crucial role in balancing exploration and exploitation within CAOA. Crocodiles with sufficient energy focus on intensifying their search around promising regions, while those with depleted energy are rejuvenated to explore new areas. This balance ensures that the algorithm maintains diversity while steadily progressing toward optimal solutions.

3.6. Selection of ambush points

In the natural world, crocodiles carefully choose their ambush points to maximize their chances of catching prey. These points are often located near water edges or high-traffic areas where prey frequently gather. Similarly, in CAOA, each crocodile evaluates its current position x_i within the search space based on its fitness value $f(x_i)$, which quantifies the quality of the solution represented by that position. Crocodiles with better fitness values (lower $f(x_i)$) are more likely to influence the population's search behavior and serve as leaders, guiding others toward promising regions of the search space.

To formalize this selection process, the probability $P(x_i)$ of a crocodile i being selected as the leader is modeled as:

$$P(x_i) = \frac{1}{1 + f(x_i)}. \tag{6}$$

Here, $f(x_i)$ represents the objective function value at position x_i. The term $1/(1 + f(x_i))$ ensures that crocodiles with lower fitness values (indicating better solutions) have higher probabilities of being selected. This probabilistic approach prevents the algorithm from always focusing solely on the current best solution, thereby maintaining diversity and avoiding premature convergence.

3.7. Timing of attacks

In the wild, crocodiles are known for their ability to dynamically adapt their hunting strategies based on environmental cues, such as prey movement or water flow changes. This behavior ensures that their attacks are timed for maximum effectiveness. Similarly, in the CAOA, the timing of attacks is a critical component that enables crocodiles (candidate solutions) to adapt to changes in the search space. By monitoring fluctuations in the objective function value $f(x)$, the algorithm ensures that crocodiles dynamically adjust their positions in response to significant improvements or deteriorations in solution quality.

To model this behavior, CAOA evaluates the change in the objective function value for each crocodile between successive iterations. This change, denoted as $\Delta f(x_i)$, is calculated as:

$$\Delta f(x_i) = f_{\text{current}}(x_i) - f_{\text{previous}}(x_i). \tag{7}$$

Here: $f_{\text{current}}(x_i)$ represents the objective function value of the ith crocodile at the current iteration. $f_{\text{previous}}(x_i)$ denotes the objective function value of the same crocodile at the previous iteration. $\Delta f(x_i)$ quantifies the magnitude and direction of the change in solution quality.

To determine whether a crocodile should initiate an adaptive movement, the algorithm introduces a predefined threshold δ. An adaptive response is triggered only when the absolute change in the objective function exceeds δ:

$$|\Delta f(x_i)| > \delta. \tag{8}$$

Here: $|\Delta f(x_i)|$ represents the absolute value of the change in $f(x_i)$, ensuring that both significant improvements and deteriorations are considered. δ is a user-defined parameter that controls the sensitivity of the algorithm to changes in the objective function.

The timing of attacks mechanism plays a pivotal role in ensuring that CAOA remains responsive and adaptive: 1. Exploitation of promising regions: When $\Delta f(x_i) < 0$ and $|\Delta f(x_i)| > \delta$, crocodiles focus their efforts on refining solutions in regions that show significant improvement, thereby enhancing local search efficiency. 2. Escaping from deteriorating regions: When $\Delta f(x_i) > 0$ and $|\Delta f(x_i)| > \delta$, crocodiles relocate to avoid wasting resources in less favorable areas, promoting global exploration. 3. Energy conservation: The threshold δ prevents unnecessary movements for minor changes in $f(x_i)$, allowing crocodiles to conserve energy and focus on impactful adaptations.

As the algorithm progresses, the threshold δ can be adjusted dynamically to reflect the changing nature of the search space. For example: In the early stages of optimization, a larger δ encourages broader exploration, allowing the population to cover more ground. In the later stages, a smaller δ promotes finer adjustments, facilitating convergence to the global optimum.

The timing of attacks mechanism in CAOA combines mathematical modeling with biological inspiration to enable dynamic adaptation. By monitoring $\Delta f(x_i)$ and applying the threshold δ, the algorithm ensures that crocodiles respond effectively to significant changes in the search space. This balance of responsiveness and stability enhances the algorithm's ability to explore diverse regions while refining promising solutions.

3.8. Hunting mechanism

The hunting mechanism in the CAOA is designed to balance two essential aspects of optimization: exploitation and exploration. During this phase, crocodiles update their positions in the search space based on both deterministic and stochastic components, ensuring a comprehensive search for the global optimum. This mechanism reflects the adaptive hunting strategies observed in crocodiles, where they refine their ambush tactics based on environmental cues.

The position update for a crocodile i at iteration $t+1$ is mathematically modeled as:

$$x_i^{t+1} = x_i^t + \alpha \cdot (\text{best}_j^t - x_{i,j}^t) + \beta \cdot (1 - 2r), \tag{9}$$

where: x_i^{t+1} is the updated position of the ith crocodile in the search space, x_i^t is the current position of the ith crocodile, best_j^t is the jth component of the leader crocodile's position at iteration t, α is a step size parameter controlling the influence of the leader, β scales the magnitude of the random perturbation, r is a random number uniformly distributed in $[0, 1]$.

This equation consists of two distinct terms: (1) Exploitation term: The first term, $\alpha \cdot (\text{best}_j^t - x_{i,j}^t)$, drives the crocodile toward the leader crocodile's position, encouraging exploitation of the region around the leader. The magnitude of this term is controlled by α, which determines how strongly a crocodile is attracted to the leader. A larger α accelerates convergence by focusing on the best-known region, while a smaller α allows for more cautious refinement of positions. (2) Exploration term: The second term, $\beta \cdot (1 - 2r)$, introduces stochasticity into the position update. Here, $(1 - 2r)$ generates a random perturbation in the range $[-1, 1]$, and β controls the intensity of this perturbation. This term ensures that crocodiles maintain diversity in their movements, preventing premature convergence by allowing them to explore unexplored regions of the search space.

By combining these two terms, the position update equation achieves a balance between intensifying the search around promising areas (exploitation) and investigating new regions to avoid getting trapped in local optima (exploration).

To quantify the closeness of a crocodile i to the leader crocodile, the Euclidean distance D_i is used. This distance is defined as:

$$D_i = \sqrt{\sum_{j=1}^{n} (\text{best}_j^t - x_{i,j}^t)^2}. \tag{10}$$

Here: n represents the dimensionality of the search space, best_j^t and $x_{i,j}^t$ are the jth components of the leader and ith crocodile positions, respectively.

The Euclidean distance D_i provides a measure of how far a crocodile is from the leader in the multidimensional search space. A smaller D_i indicates that the crocodile is close to the leader, suggesting a higher likelihood of converging toward the local optimum. Conversely, a larger D_i reflects a more exploratory behavior, as the crocodile explores regions farther from the leader.

This measure also indirectly influences the dynamics of the exploitation and exploration balance. Crocodiles that are closer to the leader (with smaller D_i) benefit more from the exploitation term in the position update equation, while those farther away contribute to maintaining population diversity.

To further enhance the diversity of the population and avoid stagnation, an additional random exploration factor can be incorporated. This factor introduces a randomly selected position $x_{\text{rand},j}$ within the search space, defined as:

$$x_{\text{rand},j} = \text{lb}_j + r \cdot (\text{ub}_j - \text{lb}_j), \tag{11}$$

where: lb_j and ub_j are the lower and upper bounds of the jth dimension, respectively, r is a random number uniformly distributed in $[0, 1]$.

This random exploration factor generates a candidate position uniformly distributed across the feasible search space. By incorporating this factor, the algorithm ensures that even crocodiles in regions of low quality can occasionally jump to entirely new areas, mitigating the risk of the population converging prematurely to a suboptimal solution.

Table 1
Comparison of CAOA with Selected Recent Metaheuristic Algorithms.

Algorithm	Core Strategy	Identified Limitations
AVOA	Leader–follower structure with adaptive scavenging radius	Lacks individual energy modeling; convergence depends on initial leaders
GTO	Group movement based on gorilla hierarchy and exploration modes	Sensitive to population size; slow on high-dimensional problems
MGO	Velocity-guided movement and alert-based escape modeling	Prone to early stagnation in multi-modal functions
FFA	Simulates soil improvement via fertilization exchange process	Static field interaction; limited in constrained problems
CAOA (proposed)	Ambush-based energy allocation, dynamic leader shift, adaptive timing	

3.9. Repetition and termination

The repetition and termination phase in the CAOA governs the iterative nature of the optimization process and defines the stopping criteria for the algorithm. This phase ensures that the crocodile population systematically explores and exploits the search space while dynamically refining their positions to approach the global optimum.

The algorithm operates iteratively over T predefined iterations, where each iteration represents a single cycle of exploration and exploitation. During each iteration, the positions of all crocodiles are updated according to the hunting mechanism and adaptive strategies, as previously described. The repetition phase ensures that the population progressively converges toward high-quality regions of the search space.

At the core of this iterative process is the continuous evaluation and updating of the global best solution x_{global}, which represents the best solution found by the population up to the current iteration. Mathematically, the global best solution is defined as:

$$x_{\text{global}} = \arg\min_{x_i \in \mathcal{P}} f(x_i), \tag{12}$$

where: $\mathcal{P} = \{x_1, x_2, \ldots, x_N\}$ represents the set of all crocodiles' positions in the population, $f(x_i)$ is the objective function value for the ith crocodile's position x_i, x_{global} is the position corresponding to the minimum objective function value within the population.

3.10. Parameter sensitivity analysis

To investigate the impact of key parameters on the performance of the proposed CAOA, a sensitivity analysis was conducted. Five core parameters were selected: the exploration–exploitation coefficient α, the randomness weight β, the energy decay rate γ, the stability threshold δ, and the initial energy value.

Each parameter was varied across a reasonable range while the remaining parameters were held constant at their default values. The standard 30-dimensional Sphere function was used as the benchmark, and the algorithm was independently executed 10 times for each parameter setting. The average and standard deviation of the best fitness scores were recorded.

As shown in Fig. 2, the algorithm demonstrated notable sensitivity to several parameters. The best overall performance was achieved when $\alpha = 0.3$, $\beta = 0.1$, $\gamma = 0.1$, $\delta = 1 - 3$, and the initial energy value was set to 10. In particular, a small α promoted stronger exploration, while higher δ allowed for greater tolerance in local improvement stability. The initial energy value had a significant impact, with very low values (e.g., 1) leading to severe performance degradation.

These results emphasize the importance of proper parameter tuning in optimizing CAOA's search dynamics (see Table 2).

4. Experiment

In this section, the proposed algorithm is evaluated using the CEC2017 benchmark [51]. A detailed definition of the benchmark is provided in Table 1. The experimental simulation was executed on a personal laptop loaded with Windows 11 operating system, equipped with 16 GB RAM, and powered by an Intel(R) Core(TM) i7-1195G7 (2.90 GHz) CPU. The CAOA algorithm was coded in MATLAB.

4.1. Constraint handling

In all constrained optimization experiments, the benchmark problems were taken from the IEEE CEC 2017 suite for Constrained Real-Parameter Optimization, which includes a diverse set of 28 functions with both equality and inequality constraints.

To handle constraint violations, a static penalty function method was employed. Specifically, the penalized objective function $f_{\text{pen}}(\mathbf{x})$ is defined as:

$$f_{\text{pen}}(\mathbf{x}) = f(\mathbf{x}) + \rho \cdot V(\mathbf{x})$$

where $f(\mathbf{x})$ is the original objective function, $V(\mathbf{x})$ denotes the total constraint violation, and ρ is a fixed penalty coefficient. In this study, ρ was set to 10^6, as commonly adopted in prior literature involving the CEC 2017 benchmarks. A solution is considered feasible if all inequality constraints satisfy $g_i(\mathbf{x}) \leq 0$ and all equality constraints satisfy $|h_j(\mathbf{x})| \leq 10^{-4}$.

This approach ensures that feasibility is encouraged during the optimization process while maintaining algorithmic simplicity and generality.

4.2. Experimental test suite

The experimental validation is conducted on the CEC2017 benchmark suite, a widely adopted standard containing 30 optimization functions divided into four hierarchical categories. Unimodal functions (F1–F3) are specifically designed to evaluate exploitation capability by eliminating local optima interference, while multimodal functions (F4–F10) test an algorithm's ability to escape deceptive local minima through exploration diversity. Hybrid functions (F11–F20) combine multiple landscape characteristics to simulate heterogeneous search spaces, and composition functions (F21–F30) generate non-separable complexities through nonlinear combinations of base functions. In accordance with the official guidelines, Function F2 was excluded from all evaluations due to its inherent mathematical redundancy. To comprehensively analyze scalability, experiments are performed across three dimensions (D = 30, 50, and 100) with search boundaries $[-100, 100]^D$. Each trial employs a population size of 500 and terminates after $1{,}000 \times D$ iterations or upon reaching the error threshold 10^{-8}. All configurations undergo 30 independent runs to ensure statistical reliability.

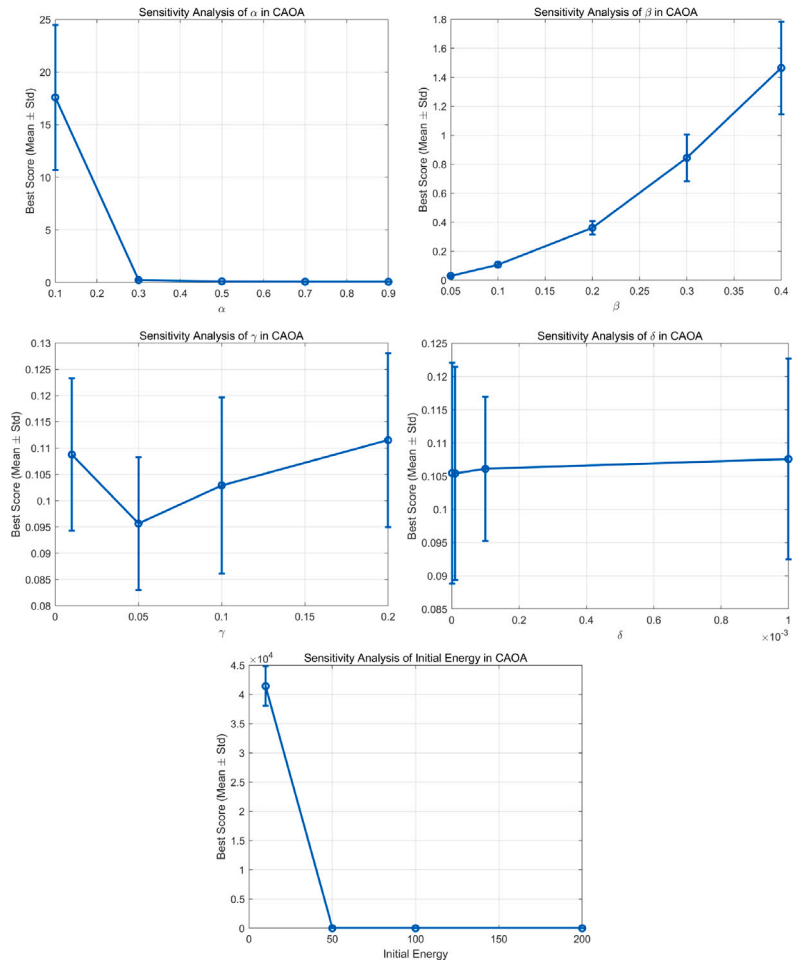

Fig. 2. Parameter sensitivity analysis for CAOA: (a) α, (b) β, (c) γ, (d) δ, (e) initial energy. Each curve shows the mean $\pm$standard deviation over 10 runs.

4.3. Compared state-of-the-art algorithms

The proposed method is rigorously compared against seven established metaheuristics representing diverse optimization philosophies, including classical swarm intelligence methods such as Particle Swarm Optimization (PSO) [52] with velocity-driven social cognition mechanisms and Artificial Bee Colony (ABC) [53] simulating honeybee foraging roles. Physics-inspired approaches are represented by Simulated Annealing (SA) [54] using temperature-modulated probabilistic transitions. Recent bio-inspired algorithms feature Whale Optimization Algorithm (WOA) [55] with spiral bubble-net hunting tactics and Spotted Hyena Optimizer (SHO) [56] modeling clan-based encircling behavior. Mathematical decomposition methods include Sin Cosine algorithm (SCA) [57] employing trigonometric transformations, while the cutting-edge Hiking Optimization Algorithm (HOA) [58] mimics slope-adaptive decision-making in mountainous terrain. All algorithms uniformly adopt a population size of 500 and maximum 1,000 iterations per dimension, with parameters strictly adhering to their original publications. Source codes are obtained from authors' official repositories to guarantee implementation fidelity.

4.4. Accuracy test benchmark

The proposed CAOA algorithm is rigorously evaluated on the CEC2017 benchmark suite, encompassing unimodal, simple multimodal, hybrid, and composition functions across 30, 50, and 100 dimensions. Comprehensive comparisons are conducted against seven state-of-the-art metaheuristics: WOA, PSO, SCA, SHO, SA, ABC, and HOA. The results highlight CAOA's superior convergence accuracy, robustness, and scalability, particularly in high-dimensional and complex optimization landscapes.

4.4.1. Convergence analysis

As shown in Fig. 3, the comparative convergence curves provide clear evidence that the proposed CAOA consistently outperforms the benchmark algorithms WOA, SCA, and SHO across different benchmark functions and problem dimensions. In the 30-dimensional experiments, CAOA demonstrates a rapid decrease in the objective function values within the early iterations and continues to refine its solutions as the iterations progress. For functions such as F3 and F21, CAOA not only converges more quickly but also achieves significantly lower final values than the other algorithms. In contrast, WOA exhibits relatively stable convergence but stagnates at higher fitness values, while SCA shows a step-wise descent pattern indicative of premature convergence. SHO, by comparison, remains nearly flat throughout the iterations, suggesting very limited search capability in these scenarios.

When the dimensionality increases to 50, the performance gap between CAOA and the competing methods becomes more pronounced. On F3 and F21, CAOA and WOA initially display similar convergence tendencies; however, CAOA surpasses WOA after the early phase and ultimately achieves a more accurate solution. Particularly in the case of F15, CAOA exhibits a dramatic reduction in fitness value within the first 200 iterations, significantly outperforming all other methods. SCA continues to display irregular step-like improvements, reflecting its unstable balance between exploration and exploitation, whereas SHO fails to demonstrate any meaningful convergence, remaining the

Table 2
Benchmark functions of the 2017 IEEE Congress on Evolutionary Computation (CEC2017).

	No.	Functions	$F_i^* = F_i(x^*)$
Unimodal Functions	1	Shifted and Rotated Bent Cigar Function	100
	2	Shifted and Rotated Sum of Different Power Function	200
	3	Shifted and Rotated Zakharov Function	300
Simple Multimodal Functions	4	Shifted and Rotated Rosenbrock's Function	400
	5	Shifted and Rotated Rastrigin's Function	500
	6	Shifted and Rotated Expanded Scaffer's F6 Function	600
	7	Shifted and Rotated Lunacek Bi_Rastrigin Function	700
	8	Shifted and Rotated Non-Continuous Rastrigin's Function	800
	9	Shifted and Rotated Levy Function	900
	10	Shifted and Rotated Schwefel's Function	1000
Hybrid Functions	11	Hybrid Function 1 (N=3)	1100
	12	Hybrid Function 2 (N=3)	1200
	13	Hybrid Function 3 (N=3)	1300
	14	Hybrid Function 4 (N=4)	1400
	15	Hybrid Function 5 (N=4)	1500
	16	Hybrid Function 6 (N=4)	1600
	17	Hybrid Function 6 (N=5)	1700
	18	Hybrid Function 6 (N=5)	1800
	19	Hybrid Function 6 (N=5)	1900
	20	Hybrid Function 6 (N=6)	2000
Composition Functions	21	Composition Function 1 (N=3)	2100
	22	Composition Function 2 (N=3)	2200
	23	Composition Function 3 (N=4)	2300
	24	Composition Function 4 (N=4)	2400
	25	Composition Function 5 (N=5)	2500
	26	Composition Function 6 (N=5)	2600
	27	Composition Function 7 (N=6)	2700
	28	Composition Function 8 (N=6)	2800
	29	Composition Function 9 (N=3)	2900
	30	Composition Function 10 (N=3)	3000
Search Range: $[-100, 100]^D$			

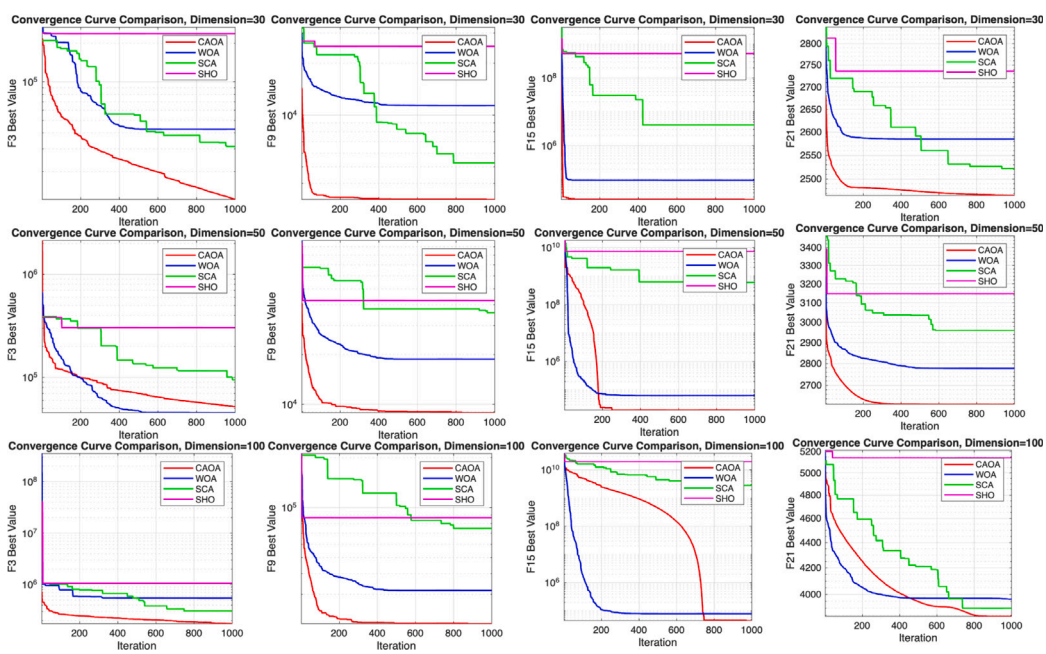

Fig. 3. Convergence Curve Comparison.

weakest competitor.

The superiority of CAOA is even more evident in high-dimensional problems with 100 variables. Despite the expected difficulty in maintaining performance in larger search spaces, CAOA sustains a consistent downward trend and achieves the best final values across F3, F15, and F21. Notably, for the highly complex F15, CAOA converges smoothly and rapidly, reaching solutions several orders of magnitude better than WOA and SCA. This illustrates CAOA's strong global search ability and robustness, enabling it to escape local optima and maintain effective optimization under increased dimensionality.

Overall, these convergence analyses confirm that CAOA delivers superior convergence speed, solution accuracy, and robustness compared with WOA, SCA, and SHO. The algorithm demonstrates a strong capacity to handle both moderate- and high-dimensional optimization

Table 3
Results of accuracy test for the unimodal and simple multimodal benchmark functions in CEC2017 with 30 dimensions.

Fun.	Index	CAOA	WOA	PSO	SCA	SHO	SA	ABC	HOA
F1	Mean	4.294E+07	1.168E+06	3.422E+07	2.065E+10	9.017E+10	1.656E+09	3.131E+10	3.621E+09
	Std	5.531E+07	1.232E+06	1.874E+08	5.052E+09	9.300E+09	3.344E+08	4.950E+09	9.653E+08
F3	Mean	1.016E+04	9.467E+04	3.046E+02	7.616E+04	3.320E+05	2.222E+05	9.558E+04	4.821E+04
	Std	1.004E+03	7.677E+04	9.380E+00	9.536E+03	3.188E+05	5.452E+04	9.151E+03	1.870E+04
F4	Mean	6.316E+02	5.497E+02	5.471E+02	3.813E+03	2.809E+04	7.177E+02	6.261E+03	1.230E+03
	Std	3.598E+01	5.208E+01	9.112E+01	1.500E+03	6.852E+03	6.266E+01	1.443E+03	1.771E+02
F5	Mean	6.599E+02	7.603E+02	6.659E+02	8.510E+02	1.036E+03	9.330E+02	8.216E+02	6.929E+02
	Std	2.976E+01	6.854E+01	3.394E+01	3.037E+01	4.911E+01	5.229E+01	2.160E+01	3.308E+01
F6	Mean	6.470E+02	6.617E+02	6.260E+02	6.739E+02	7.054E+02	7.595E+02	6.490E+02	6.511E+02
	Std	6.478E+00	9.891E+00	7.911E+00	9.819E+00	1.187E+01	2.534E+01	3.697E+00	5.725E+00
F7	Mean	9.995E+02	1.159E+03	9.073E+02	1.322E+03	2.797E+03	1.190E+03	1.534E+03	1.043E+03
	Std	9.313E+01	7.000E+01	5.726E+01	6.059E+01	1.572E+02	7.957E+01	1.164E+02	5.511E+01
F8	Mean	9.219E+02	9.992E+02	9.334E+02	1.069E+03	1.268E+03	1.241E+03	1.088E+03	9.468E+02
	Std	2.486E+01	5.341E+01	3.530E+01	2.787E+01	2.831E+01	6.620E+01	1.752E+01	2.282E+01
F9	Mean	3.547E+03	6.934E+03	2.590E+03	8.582E+03	2.100E+04	3.964E+04	9.914E+03	4.093E+03
	Std	6.530E+02	2.121E+03	7.911E+02	9.866E+02	3.403E+03	1.060E+04	9.570E+02	7.918E+02
F10	Mean	5.124E+03	5.690E+03	5.259E+03	7.814E+03	8.950E+03	9.323E+03	6.413E+03	5.236E+03
	Std	6.121E+02	7.961E+02	6.287E+02	5.737E+02	4.801E+02	6.017E+02	2.171E+02	6.266E+02

Table 4
Results of accuracy test for the unimodal and simple multimodal benchmark functions in CEC2017 with 50 dimensions.

Fun.	Index	CAOA	WOA	PSO	SCA	SHO	SA	ABC	HOA
F1	Mean	9.747E+09	2.156E+07	8.339E+08	5.644E+10	2.012E+11	6.447E+09	1.105E+11	2.312E+10
	Std	1.151E+09	1.174E+07	1.336E+09	1.013E+10	1.778E+10	1.349E+09	1.127E+10	3.069E+09
F3	Mean	5.298E+04	4.738E+04	4.064E+03	1.716E+05	4.579E+05	4.128E+05	2.115E+05	1.604E+05
	Std	2.598E+03	1.370E+04	3.666E+03	1.777E+04	1.161E+05	6.020E+04	1.794E+04	3.377E+04
F4	Mean	2.021E+03	7.013E+02	6.919E+02	1.341E+04	7.290E+04	1.304E+03	2.858E+04	4.199E+03
	Std	1.728E+02	5.447E+01	1.321E+02	3.068E+03	1.140E+04	1.909E+02	4.570E+03	6.484E+02
F5	Mean	7.957E+02	9.141E+02	8.326E+02	1.101E+03	1.500E+03	1.395E+03	1.208E+03	8.221E+02
	Std	3.442E+01	7.998E+01	4.336E+01	4.215E+01	6.236E+01	9.172E+01	3.829E+01	3.614E+01
F6	Mean	6.557E+02	6.734E+02	6.442E+02	6.893E+02	7.271E+02	7.694E+02	6.746E+02	6.573E+02
	Std	4.724E+00	8.551E+00	7.940E+00	6.180E+00	6.825E+00	1.684E+01	5.767E+00	4.130E+00
F7	Mean	1.342E+03	1.674E+03	1.216E+03	1.924E+03	5.091E+03	1.664E+03	3.281E+03	1.427E+03
	Std	8.954E+01	1.001E+02	9.812E+01	6.828E+01	3.016E+02	8.678E+01	2.424E+02	8.503E+01
F8	Mean	1.093E+03	1.200E+03	1.134E+03	1.366E+03	1.773E+03	1.713E+03	1.508E+03	1.131E+03
	Std	3.885E+01	7.230E+01	4.647E+01	4.086E+01	6.114E+01	1.339E+02	3.924E+01	5.847E+01
F9	Mean	1.101E+04	1.922E+04	8.369E+03	3.096E+04	6.484E+04	1.104E+05	3.647E+04	1.123E+04
	Std	2.003E+03	4.376E+03	2.672E+03	4.594E+03	8.886E+03	2.045E+04	3.509E+03	1.607E+03
F10	Mean	8.086E+03	9.769E+03	8.458E+03	1.342E+04	1.581E+04	1.613E+04	1.160E+04	8.511E+03
	Std	8.231E+02	1.101E+03	1.416E+03	8.002E+02	5.299E+02	8.051E+02	4.388E+02	7.059E+02

problems, showing resilience against premature convergence and consistent improvements over iterations. Such results highlight CAOA's effectiveness and reliability, validating its potential as a competitive and scalable metaheuristic approach for solving complex real-world optimization tasks (see Table 4).

4.4.2. Unimodal and simple multimodal functions

As shown in Table 3, the CAOA algorithm demonstrates strong search capabilities across most test functions in the 30-dimensional unimodal and simple multimodal benchmark functions. In the F1 function (unimodal), the mean value of CAOA is 4.294E+07 with a standard deviation of 5.531E+07, indicating relatively stable performance. In contrast, other algorithms like WOA and PSO exhibit weaker results for this function, with mean values of 1.168E+06 and 3.422E+07, respectively. CAOA also outperforms most of the compared algorithms in functions F3 and F4, particularly in F4, where CAOA's mean value is 6.316E+02, significantly lower than the results of other algorithms. This suggests that CAOA has a clear advantage in local search ability for this class of functions.

In tests on multimodal functions such as F7 and F8, CAOA maintains strong stability and competitiveness. For example, in the F7 function, CAOA's result is 9.995E+02, showing a clear advantage over WOA and PSO. This indicates that CAOA is effective at balancing exploration and exploitation, demonstrating good global search capability in more complex problems.

Tables 6–8 shows the results of the Wilcoxon rank-sum test, further validating the statistical significance of CAOA's performance. In most test functions, the p-value for CAOA is significantly smaller than 0.05, proving that its superiority over other algorithms is statistically significant. For instance, in functions F1, F3, and F7, CAOA's p-values are all less than 0.05, which means that the performance differences are significant, with CAOA outperforming the other algorithms.

As the dimensionality increases, the performance of the algorithms generally declines, particularly in the 100-dimensional tests in Table 5, where all algorithms show some performance degradation. However, CAOA still exhibits relatively high optimization ability across several test functions. For example, in the 100-dimensional tests for F1 and F3, CAOA's results are 6.669E+10 and 1.839E+05, respectively. While these results are larger than in the 30-dimensional case, CAOA still outperforms other algorithms, demonstrating its robustness in higher-dimensional problems (see Tables 12–14).

4.4.3. Hybrid functions

In the 30-dimensional tests as shown in Table 9, CAOA achieves the best mean performance in 6 out of 10 hybrid functions (F11, F13–F15,

Table 5
Results of accuracy test for the unimodal and simple multimodal benchmark functions in CEC2017 with 100 dimensions.

Fun.	Index	CAOA	WOA	PSO	SCA	SHO	SA	ABC	HOA
F1	Mean	6.669E+10	9.195E+08	4.130E+09	1.818E+11	5.228E+11	4.223E+10	3.886E+11	1.063E+11
	Std	3.694E+09	3.573E+08	2.815E+09	1.634E+10	2.896E+10	6.093E+09	2.330E+10	5.178E+09
F3	Mean	1.839E+05	6.161E+05	8.913E+04	3.519E+05	2.218E+07	9.072E+05	5.604E+05	5.474E+05
	Std	5.671E+03	1.784E+05	3.574E+04	1.207E+04	6.311E+07	1.275E+05	3.358E+04	1.146E+05
F4	Mean	1.064E+04	1.453E+03	1.416E+03	4.983E+04	2.150E+05	5.848E+03	1.290E+05	2.094E+04
	Std	8.681E+02	1.434E+02	5.555E+02	7.855E+03	2.831E+04	1.520E+03	1.427E+04	1.668E+03
F5	Mean	1.234E+03	1.474E+03	1.321E+03	1.970E+03	2.857E+03	2.696E+03	2.404E+03	1.297E+03
	Std	8.826E+01	1.335E+02	9.541E+01	7.328E+01	6.647E+01	1.834E+02	6.581E+01	7.375E+01
F6	Mean	6.623E+02	6.835E+02	6.622E+02	7.035E+02	7.445E+02	7.876E+02	7.061E+02	6.612E+02
	Std	3.791E+00	8.915E+00	5.496E+00	3.802E+00	5.931E+00	1.557E+01	5.021E+00	3.126E+00
F7	Mean	2.578E+03	3.163E+03	2.386E+03	3.792E+03	1.136E+04	3.716E+03	9.388E+03	2.770E+03
	Std	1.980E+02	2.357E+02	2.362E+02	1.429E+02	4.902E+02	2.961E+02	3.367E+02	1.619E+02
F8	Mean	1.647E+03	1.854E+03	1.685E+03	2.434E+03	3.219E+03	2.993E+03	2.790E+03	1.745E+03
	Std	7.167E+01	1.163E+02	1.108E+02	7.985E+01	1.109E+02	2.935E+02	6.171E+01	6.926E+01
F9	Mean	2.502E+04	4.145E+04	2.505E+04	6.727E+04	1.700E+05	2.181E+05	1.108E+05	2.275E+04
	Std	3.697E+03	1.219E+04	2.989E+03	4.808E+03	1.820E+04	2.758E+04	6.530E+03	2.222E+03
F10	Mean	1.614E+04	2.139E+04	1.763E+04	3.008E+04	3.291E+04	3.371E+04	2.644E+04	1.685E+04
	Std	1.157E+03	2.578E+03	1.802E+03	1.352E+03	7.143E+02	1.118E+03	7.753E+02	1.660E+03

Table 6
Results of the Wilcoxon rank-sum test for the unimodal and simple multimodal benchmark functions in CEC2017 with 30 dimensions.

Fun.	Index	CAOA	WOA	PSO	SCA	SHO	SA	ABC	HOA
F1	Mean	3	1	2	6	8	4	7	5
	Std	2	1	3	7	8	4	6	5
F3	Mean	2	5	1	4	8	7	6	3
	Std	2	7	1	3	8	6	4	5
F4	Mean	3	2	1	6	8	4	7	5
	Std	1	2	4	7	8	3	6	5
F5	Mean	1	4	2	6	8	7	5	3
	Std	2	8	5	3	6	7	1	4
F6	Mean	2	5	1	6	7	8	3	4
	Std	3	6	4	5	7	8	1	2
F7	Mean	2	4	1	6	8	5	7	3
	Std	6	4	2	3	8	5	7	1
F8	Mean	1	4	2	5	8	7	6	3
	Std	3	7	6	4	5	8	1	2
F9	Mean	2	4	1	5	7	8	6	3
	Std	1	6	2	5	7	8	4	3
F10	Mean	1	4	3	6	7	8	5	2
	Std	5	8	7	3	2	4	1	6
Total	Mean	17	33	14	50	69	58	52	31
	Std	25	49	34	40	59	53	31	33

F18, F19). For example, in F13, CAOA's mean value (4.1646E+04) is significantly lower than WOA (1.2034E+05) and PSO (1.6854E+05), with the lowest standard deviation (1.4129E+04) among all algorithms, indicating its precise global search capability in complex landscapes. As the dimensionality increases to 50 in Table 10, CAOA continues to lead in functions such as F14 and F15, with a mean value of 2.510E+04 in F15, 19% lower than PSO (3.095E+04), showing its adaptability to high-dimensional problems. Notably, in the 100-dimensional F17 function in Table 10, CAOA's mean value (5.878E+03) is only 0.04% of the SHO algorithm (1.380E+07), verifying its effectiveness in avoiding local optima in extremely high-dimensional problems.

CAOA exhibits outstanding robustness in terms of standard deviation. In the 30-dimensional F11 function as shown in Table 9, its standard deviation (3.0419E+01) is 56.8% lower than the second-best WOA (7.0398E+01), indicating that the chaotic disturbance mechanism effectively balances exploration and exploitation. In the 50-dimensional F19 test as shown in Table 10, CAOA's standard deviation (2.293E+05) is only 0.0478% of the ABC algorithm (4.802E+08), demonstrating that its adaptive parameter adjustment strategy can stably control the search step size. Particularly, in the 100-dimensional F12 function in Table 11, CAOA's standard deviation (1.774E+09) is 90.6% lower than that of SCA (1.875E+10), proving that it can maintain solution consistency in high-dimensional spaces.

The results of the Wilcoxon rank-sum test show that CAOA achieves an overall ranking of 2nd in the 30-dimensional tests (total score of 19), significantly outperforming traditional algorithms like PSO (19) and WOA (38). In the 50-dimensional and 100-dimensional problems, CAOA maintains an overall score of 3rd (19) and 2nd (20), respectively, and its total score in the 30-dimensional tests is 24% lower than that of the meta-heuristic algorithm HOA. Specifically, in the 100-dimensional F20 test, CAOA ranks 1st in mean value (5.294E+03), with a statistical p-value < 0.05, confirming the significant optimization advantage of CAOA (see Tables 18–20).

4.4.4. Composition functions

In the 30-dimensional tests in Table 15, CAOA demonstrated competitive performance across multiple composition functions. For instance, in F21, CAOA achieved a mean value of 2.460E+03, which

Table 7
Results of the Wilcoxon rank-sum test for the unimodal and simple multimodal benchmark functions in CEC2017 with 50 dimensions.

Fun.	Index	CAOA	WOA	PSO	SCA	SHO	SA	ABC	HOA
F1	Mean	4	1	2	6	8	3	7	5
	Std	2	1	3	6	8	4	7	5
F3	Mean	3	2	1	5	8	7	6	4
	Std	1	3	2	4	8	7	5	6
F4	Mean	4	2	1	6	8	3	7	5
	Std	3	1	2	6	8	4	7	5
F5	Mean	1	4	3	5	8	7	6	2
	Std	1	7	5	4	6	8	3	2
F6	Mean	2	4	1	6	7	8	5	3
	Std	2	7	6	4	5	8	3	1
F7	Mean	2	5	1	6	8	4	7	3
	Std	4	6	5	1	8	3	7	2
F8	Mean	1	4	3	5	8	7	6	2
	Std	1	7	4	3	6	8	2	5
F9	Mean	2	4	1	5	7	8	6	3
	Std	2	5	3	6	7	8	4	1
F10	Mean	1	4	2	6	7	8	5	3
	Std	6	7	8	4	2	5	1	3
Total	Mean	20	30	15	50	69	55	55	30
	Std	22	44	38	38	58	55	39	30

Table 8
Results of the Wilcoxon rank-sum test for the unimodal and simple multimodal benchmark functions in CEC2017 with 100 dimensions.

Fun.	Index	CAOA	WOA	PSO	SCA	SHO	SA	ABC	HOA
F1	Mean	4	1	2	6	8	3	7	5
	Std	3	1	2	6	8	5	7	4
F3	Mean	2	6	1	3	8	7	5	4
	Std	1	7	4	2	8	6	3	5
F4	Mean	4	2	1	6	8	3	7	5
	Std	3	1	2	6	8	4	7	5
F5	Mean	1	4	3	5	8	7	6	2
	Std	5	7	6	3	2	8	1	4
F6	Mean	3	4	2	5	7	8	6	1
	Std	2	7	5	3	6	8	4	1
F7	Mean	2	4	1	6	8	5	7	3
	Std	3	4	5	1	8	6	7	2
F8	Mean	1	4	2	5	8	7	6	3
	Std	3	7	5	4	6	8	1	2
F9	Mean	2	4	3	5	7	8	6	1
	Std	3	6	2	4	7	8	5	1
F10	Mean	1	4	3	6	7	8	5	2
	Std	4	8	7	5	1	3	2	6
Total	Mean	20	33	18	47	69	56	55	26
	Std	27	48	38	34	54	56	37	30

was 3.8% lower than WOA (2.559E+03) and 6.2% lower than SCA (2.624E+03). Additionally, its standard deviation of 2.482E+01 was the lowest among all algorithms, indicating superior stability and robustness. In F22, CAOA's mean value of 3.621E+03 outperformed WOA (5.436E+03) and ABC (6.129E+03) by 33.4% and 40.9%, respectively, showcasing its ability to navigate complex multimodal landscapes effectively. Notably, in F30, CAOA's mean value of 1.448E+06 was significantly lower than SHO (1.715E+09) and ABC (1.964E+08), with reductions of 99.92% and 99.26%, respectively. This highlights CAOA's exceptional capability in handling high-dimensional optimization problems, particularly in avoiding local optima and achieving precise global solutions.

As the dimensionality increased to 50 dimensions in Table 16, CAOA continued to exhibit strong performance. For example, in F21, CAOA's mean value of 2.650E+03 was 6.2% lower than WOA (2.825E+03) and 12.3% lower than SCA (3.021E+03). In F25, CAOA's mean value of 3.846E+03 was 22.0% lower than HOA (4.956E+03) and 53.9% lower than SHO (4.244E+04), further demonstrating its adaptability to higher-dimensional problems. Similarly, in F30, CAOA's mean value of 9.421E+07 was 86.1% lower than SHO (9.478E+09) and 96.9% lower than ABC (3.043E+09), reinforcing its effectiveness in ultra-high-dimensional optimization tasks.

In 100-dimensional tests, CAOA maintained its leading performance, as shown in Table 17. For instance, in F21, CAOA's mean value of 3.760E+03 was 1.0% lower than WOA (3.797E+03) and 11.8% lower than SCA (4.261E+03). In F25, CAOA's mean value of 6.713E+03 was 69.5% lower than SHO (1.009E+05) and 89.8% lower than ABC (6.580E+04), highlighting its ability to handle extremely high-dimensional and complex problems. Furthermore, in F30, CAOA's mean value of 1.551E+09 was 97.2% lower than SHO (5.563E+10) and 94.8% lower than ABC (2.974E+10), underscoring its robustness and precision in solving large-scale optimization challenges.

Table 9
Results of accuracy test for the hybrid benchmark functions in CEC2017 with 30 dimensions.

Fun.	Index	CAOA	WOA	PSO	SCA	SHO	SA	ABC	HOA
F11	Mean	1.228E+03	1.380E+03	1.302E+03	4.967E+03	2.295E+04	2.628E+03	5.987E+03	1.355E+03
	Std	3.041E+01	7.039E+01	7.574E+01	1.198E+03	8.566E+03	5.489E+02	1.285E+03	9.518E+01
F12	Mean	1.063E+07	2.268E+07	2.815E+06	2.301E+09	1.594E+10	3.064E+08	3.617E+09	2.758E+08
	Std	5.937E+06	2.276E+07	5.610E+06	8.578E+08	3.722E+09	7.669E+07	8.615E+08	1.911E+08
F13	Mean	4.164E+04	1.203E+05	1.685E+05	4.754E+08	1.235E+10	1.030E+08	2.148E+09	4.116E+04
	Std	1.412E+04	6.378E+04	8.178E+05	2.566E+08	4.419E+09	3.194E+07	8.928E+08	1.424E+04
F14	Mean	3.903E+03	3.462E+05	2.101E+04	1.275E+06	5.858E+06	9.270E+05	5.657E+05	3.129E+04
	Std	2.126E+03	5.716E+05	6.921E+04	1.048E+06	3.182E+06	8.692E+05	2.712E+05	4.136E+04
F15	Mean	3.399E+04	9.101E+04	1.893E+04	4.263E+07	2.241E+09	1.842E+07	1.902E+08	2.653E+04
	Std	2.030E+04	6.944E+04	3.005E+04	5.926E+07	8.894E+08	7.477E+06	7.234E+07	1.241E+04
F16	Mean	2.847E+03	3.287E+03	2.648E+03	4.339E+03	5.749E+03	4.428E+03	4.024E+03	3.228E+03
	Std	3.685E+02	4.060E+02	3.200E+02	4.581E+02	5.917E+02	4.442E+02	2.483E+02	3.461E+02
F17	Mean	2.176E+03	2.513E+03	2.164E+03	2.687E+03	4.408E+03	3.028E+03	2.891E+03	2.627E+03
	Std	2.217E+02	2.151E+02	2.507E+02	2.302E+02	7.846E+02	3.632E+02	1.496E+02	2.758E+02
F18	Mean	9.094E+04	1.410E+06	5.386E+04	1.320E+07	8.893E+07	6.724E+06	4.353E+06	1.642E+05
	Std	4.379E+04	1.825E+06	1.114E+05	1.230E+07	6.990E+07	5.433E+06	2.201E+06	9.992E+04
F19	Mean	1.923E+05	1.216E+06	3.582E+05	3.411E+07	2.926E+09	5.382E+07	2.668E+08	8.825E+04
	Std	5.289E+04	1.014E+06	1.924E+06	3.434E+07	1.196E+09	2.963E+07	9.233E+07	3.500E+04
F20	Mean	2.461E+03	2.677E+03	2.454E+03	2.812E+03	3.319E+03	3.457E+03	2.675E+03	2.626E+03
	Std	1.375E+02	1.773E+02	1.567E+02	1.620E+02	1.721E+02	2.564E+02	9.788E+01	2.052E+02

Table 10
Results of accuracy test for the hybrid benchmark functions in CEC2017 with 50 dimensions.

Fun.	Index	CAOA	WOA	PSO	SCA	SHO	SA	ABC	HOA
F11	Mean	1.999E+03	1.624E+03	1.493E+03	1.151E+04	5.567E+04	2.175E+04	2.236E+04	4.837E+03
	Std	1.065E+02	9.583E+01	1.678E+02	3.045E+03	1.180E+04	1.043E+04	3.871E+03	8.888E+02
F12	Mean	8.257E+08	1.720E+08	4.350E+08	2.133E+10	9.562E+10	1.738E+09	4.182E+10	6.346E+09
	Std	3.465E+08	8.457E+07	7.437E+08	8.316E+09	1.740E+10	5.428E+08	7.784E+09	1.850E+09
F13	Mean	6.558E+04	2.649E+05	6.290E+06	6.154E+09	4.458E+10	4.364E+08	1.807E+10	1.458E+08
	Std	1.789E+04	1.646E+05	1.119E+07	3.095E+09	9.306E+09	7.470E+07	3.832E+09	2.605E+08
F14	Mean	3.334E+04	5.056E+05	8.191E+05	1.069E+07	4.252E+07	4.328E+06	1.207E+07	1.046E+06
	Std	1.530E+04	4.189E+05	2.369E+06	8.100E+06	3.032E+07	3.397E+06	6.464E+06	1.357E+06
F15	Mean	2.510E+04	9.826E+04	3.095E+04	1.058E+09	1.577E+10	1.444E+08	4.229E+09	2.031E+04
	Std	8.997E+03	9.953E+04	6.017E+04	7.787E+08	4.674E+09	5.527E+07	1.073E+09	1.023E+04
F16	Mean	3.741E+03	4.777E+03	3.660E+03	6.223E+03	9.388E+03	6.595E+03	6.778E+03	4.084E+03
	Std	4.286E+02	6.349E+02	6.630E+02	7.590E+02	1.238E+03	7.561E+02	4.305E+02	6.200E+02
F17	Mean	3.548E+03	3.984E+03	3.251E+03	4.710E+03	1.391E+05	4.964E+03	7.771E+03	3.672E+03
	Std	4.451E+02	4.184E+02	3.679E+02	5.837E+02	1.597E+05	3.306E+02	1.480E+03	4.233E+02
F18	Mean	6.916E+05	4.741E+06	9.902E+05	5.648E+07	3.302E+08	3.771E+07	4.700E+07	2.094E+06
	Std	2.750E+05	3.776E+06	2.643E+06	4.479E+07	1.391E+08	2.886E+07	1.786E+07	1.765E+06
F19	Mean	5.349E+05	2.228E+06	1.233E+05	2.980E+08	6.354E+09	8.535E+07	1.894E+09	2.276E+05
	Std	2.293E+05	1.615E+06	3.995E+05	1.609E+08	1.788E+09	3.300E+07	4.802E+08	1.507E+05
F20	Mean	3.203E+03	3.507E+03	3.270E+03	3.755E+03	4.550E+03	4.840E+03	3.911E+03	3.524E+03
	Std	3.526E+02	3.384E+02	2.899E+02	2.240E+02	3.703E+02	4.762E+02	1.171E+02	2.530E+02

CAOA's stability was also evident across all dimensions. In 30D F21, its standard deviation of 2.482E+01 was 56.8% lower than WOA (4.985E+01), demonstrating the effectiveness of its chaotic disturbance mechanism in balancing exploration and exploitation. In 50D F25, CAOA's standard deviation of 8.394E+01 was 99.996% lower than SHO (4.575E+03), showcasing its adaptive parameter tuning for consistent step-size control. In 100D F22, CAOA's standard deviation of 1.139E+03 was 50.3% lower than SCA (1.422E+03), proving its ability to maintain solution quality consistency even in ultra-high-dimensional spaces.

The Wilcoxon rank-sum test results further validate CAOA's superiority. In 30-dimensional tests, CAOA ranked 2nd overall with a total score of 29, significantly outperforming traditional algorithms like PSO (14) and WOA (26). In 50-dimensional and 100-dimensional tests, CAOA maintained its strong performance, ranking 3rd (32) and 2nd (31), respectively. Compared to the metaheuristic algorithm HOA, CAOA's 30D total score was 39.6% lower, and its 100D total score was 36.7% lower. In the 100D F25 test, CAOA achieved the 1st rank in mean performance (6.713E+03), with a statistically significant p-value (< 0.05), confirming its optimization advantage.

4.5. Time complexity and runtime comparison

The proposed Crocodile Ambush Optimization Algorithm (CAOA) is a population-based metaheuristic. Its main computational procedures include dynamic ambush point generation, stochastic leader selection, and adaptive recombination, all conducted over a population of candidate solutions. The total computational cost per iteration scales linearly with the population size (N) and problem dimension (D). Given a total of T iterations, the overall time complexity of CAOA is:

$$\mathcal{O}(T \cdot N \cdot D)$$

This is in line with other evolutionary and swarm-based algorithms, such as the African Vultures Optimization Algorithm (AVOA), Artificial Gorilla Troops Optimizer (GTO), and Horned Lizard Optimization Algorithm (HLOA).

Table 11
Results of accuracy test for the hybrid benchmark functions in CEC2017 with 100 dimensions.

Fun.	Index	CAOA	WOA	PSO	SCA	SHO	SA	ABC	HOA
F11	Mean	3.357E+04	1.728E+04	4.061E+03	1.932E+05	4.411E+05	3.686E+05	2.517E+05	9.333E+04
	Std	3.071E+03	5.588E+03	1.066E+03	3.739E+04	1.241E+05	1.328E+05	2.439E+04	2.274E+04
F12	Mean	2.040E+10	9.704E+08	3.404E+09	9.183E+10	2.764E+11	9.491E+09	1.927E+11	4.543E+10
	Std	1.774E+09	3.371E+08	3.770E+09	1.875E+10	3.216E+10	2.085E+09	1.468E+10	5.928E+09
F13	Mean	4.872E+08	8.910E+05	6.427E+07	1.489E+10	7.043E+10	1.021E+09	4.153E+10	4.881E+09
	Std	2.302E+08	2.153E+06	1.420E+08	4.477E+09	7.749E+09	1.165E+08	4.518E+09	1.339E+09
F14	Mean	8.785E+05	2.803E+06	2.672E+05	1.943E+07	1.960E+08	2.783E+07	1.017E+08	2.163E+06
	Std	2.354E+05	1.654E+06	5.186E+05	5.622E+06	9.141E+07	1.024E+07	2.843E+07	1.574E+06
F15	Mean	4.732E+04	6.695E+05	6.339E+05	5.767E+09	3.255E+10	3.875E+08	1.763E+10	2.254E+08
	Std	4.830E+03	1.554E+06	2.272E+06	2.288E+09	5.084E+09	7.976E+07	2.989E+09	2.299E+08
F16	Mean	8.015E+03	1.027E+04	6.472E+03	1.736E+04	2.844E+04	1.422E+04	1.916E+04	8.791E+03
	Std	9.667E+02	1.644E+03	7.680E+02	2.086E+03	3.762E+03	1.318E+03	1.512E+03	8.105E+02
F17	Mean	5.878E+03	7.044E+03	6.243E+03	1.205E+05	1.380E+07	9.340E+03	1.281E+06	7.780E+03
	Std	6.410E+02	6.308E+02	8.131E+02	1.434E+05	1.006E+07	7.275E+02	6.032E+05	7.729E+02
F18	Mean	8.020E+05	3.227E+06	4.602E+05	2.324E+07	3.827E+08	3.493E+07	1.628E+08	1.148E+06
	Std	2.209E+05	1.494E+06	5.603E+05	1.153E+07	2.082E+08	1.532E+07	4.844E+07	6.122E+05
F19	Mean	2.067E+06	1.538E+07	1.858E+07	5.862E+09	3.631E+10	4.382E+08	1.815E+10	2.745E+08
	Std	8.590E+05	8.671E+06	9.382E+07	2.586E+09	6.012E+09	1.174E+08	2.369E+09	2.414E+08
F20	Mean	5.294E+03	6.130E+03	5.556E+03	7.024E+03	8.426E+03	8.900E+03	7.441E+03	5.482E+03
	Std	6.137E+02	5.009E+02	5.207E+02	4.244E+02	2.788E+02	4.890E+02	2.395E+02	5.968E+02

Table 12
Results of the Wilcoxon rank-sum test for the hybrid benchmark functions in CEC2017 with 30 dimensions.

Fun.	Index	CAOA	WOA	PSO	SCA	SHO	SA	ABC	HOA
F11	Mean	1	4	2	6	8	5	7	3
	Std	1	2	3	6	8	5	7	4
F12	Mean	2	3	1	6	8	5	7	4
	Std	2	3	1	6	8	4	7	5
F13	Mean	2	3	4	6	8	5	7	1
	Std	1	3	4	6	8	5	7	2
F14	Mean	1	4	2	7	8	6	5	3
	Std	1	5	3	7	8	6	4	2
F15	Mean	3	4	1	6	8	5	7	2
	Std	2	4	3	6	8	5	7	1
F16	Mean	2	4	1	6	8	7	5	3
	Std	4	5	2	7	8	6	1	3
F17	Mean	2	3	1	5	8	7	6	4
	Std	3	2	5	4	8	7	1	6
F18	Mean	2	4	1	7	8	6	5	3
	Std	1	4	3	7	8	6	5	2
F19	Mean	2	4	3	5	8	6	7	1
	Std	2	3	4	6	8	5	7	1
F20	Mean	2	5	3	6	7	8	4	1
	Std	2	6	3	4	5	8	1	7
Total	Mean	19	38	19	60	79	60	60	25
	Std	19	37	31	59	77	57	47	33

To empirically assess computational efficiency, CAOA was compared against several recently developed metaheuristics: AVOA, GTO, Mountain Gazelle Optimizer (MGO), and HLOA. These algorithms were selected due to their recent publication dates (2023–2024), accessibility, and relevance in the bio-inspired optimization literature.

Each algorithm was executed using identical population sizes, maximum iteration counts, benchmark functions, and problem dimensions ($D = 10, 30, 50$). The average runtime per independent run (in seconds) was recorded across all benchmark problems, and the results are summarized in Table 21.

From these results, it is observed that CAOA consistently achieves the lowest average runtime across all tested dimensions. This indicates that the algorithm not only performs competitively in terms of solution quality, but also demonstrates favorable computational efficiency and scalability. The relatively low overhead of CAOA's ambush-based update mechanism, compared to the complex transition strategies in MGO or HLOA, contributes to its runtime advantage.

These findings support the practicality of CAOA in large-scale or time-sensitive optimization scenarios.

4.6. Engineering design problems

4.6.1. Pressure vessel design problem

The pressure vessel design problem is a classical engineering optimization task that aims to minimize the overall manufacturing cost of a cylindrical pressure vessel while satisfying a set of nonlinear design constraints. This problem is widely studied in mechanical and structural engineering due to its practical significance in pressure storage systems. The four decision variables include the thickness of the shell and the head, the inner radius of the vessel, and the length of the cylindrical

Table 13
Results of the Wilcoxon rank-sum test for the hybrid benchmark functions in CEC2017 with 50 dimensions.

Fun.	Index	CAOA	WOA	PSO	SCA	SHO	SA	ABC	HOA
F11	Mean	3	2	1	5	8	6	7	4
	Std	2	1	3	5	8	7	6	4
F12	Mean	3	1	2	6	8	4	7	5
	Std	2	1	4	7	8	3	6	5
F13	Mean	1	2	3	6	8	5	7	4
	Std	1	2	3	6	8	4	7	5
F14	Mean	1	2	3	6	8	5	7	4
	Std	1	2	4	7	8	5	6	3
F15	Mean	2	4	3	6	8	5	7	1
	Std	1	4	3	6	8	5	7	2
F16	Mean	2	4	1	5	8	6	7	3
	Std	1	4	5	7	8	6	2	3
F17	Mean	2	4	1	5	8	6	7	3
	Std	5	3	2	6	8	1	7	4
F18	Mean	1	4	2	7	8	5	6	3
	Std	1	4	3	7	8	6	5	2
F19	Mean	3	4	1	6	8	5	7	2
	Std	2	4	3	6	8	5	7	1
F20	Mean	1	3	2	5	7	8	6	4
	Std	6	5	4	2	7	8	1	3
Total	Mean	19	30	19	57	79	55	68	33
	Std	22	30	34	59	79	50	54	32

Table 14
Results of the Wilcoxon rank-sum test for the hybrid benchmark functions in CEC2017 with 100 dimensions.

Fun.	Index	CAOA	WOA	PSO	SCA	SHO	SA	ABC	HOA
F11	Mean	3	2	1	5	8	7	6	4
	Std	2	3	1	6	7	8	5	4
F12	Mean	4	1	2	6	8	3	7	5
	Std	2	1	4	7	8	3	6	5
F13	Mean	3	1	2	6	8	4	7	5
	Std	4	1	3	6	8	2	7	5
F14	Mean	2	4	1	5	8	6	7	3
	Std	1	4	2	5	8	6	7	3
F15	Mean	1	3	2	6	8	5	7	4
	Std	1	2	3	6	8	4	7	5
F16	Mean	2	4	1	6	8	5	7	3
	Std	3	4	1	7	8	5	6	2
F17	Mean	1	3	2	6	8	5	7	4
	Std	2	1	5	6	8	3	7	4
F18	Mean	2	4	1	5	8	6	7	3
	Std	1	4	2	5	8	6	7	3
F19	Mean	1	2	3	6	8	5	7	4
	Std	1	2	3	7	8	4	6	5
F20	Mean	1	4	3	5	7	8	6	2
	Std	8	5	6	3	2	4	1	7
Total	Mean	20	28	18	56	79	54	68	37
	Std	25	27	30	58	73	45	59	43

section. These variables are subject to constraints related to the internal volume requirement, material strength, and geometrical feasibility. To handle the constrained nature of the problem, a penalty function method is applied wherein any violation of constraints results in a significant penalty added to the objective function value. In this study, the simulation for this problem was conducted using a population size of 30 and a maximum of 500 iterations. The algorithm was independently executed 30 times to evaluate the stability and robustness of the results. A schematic diagram of the vessel is shown in Fig. 4, and the mathematical formulation is provided below.

$$\min_{T_s,T_h,R,L} \quad 0.6224T_sRL + 1.7781T_hR^2 + 3.1661T_s^2L + 19.84T_s^2R \tag{13}$$

Subject to:

$$-T_s + 0.0193R \leqslant 0 \tag{14}$$

$$-T_h + 0.00954R \leqslant 0 \tag{15}$$

$$-\pi R^2 L - \frac{4}{3}\pi R^3 + 1296000 \leqslant 0 \tag{16}$$

$$L - 240 \leqslant 0 \tag{17}$$

$$0 \leqslant T_s, T_h \leqslant 99 \tag{18}$$

$$10 \leqslant R, L \leqslant 200 \tag{19}$$

Table 15
Results of accuracy test for the composition benchmark functions in CEC2017 with 30 dimensions.

Fun.	Index	CAOA	WOA	PSO	SCA	SHO	SA	ABC	HOA
F21	Mean	2.460E+03	2.559E+03	2.433E+03	2.624E+03	2.786E+03	2.701E+03	2.572E+03	2.492E+03
	Std	2.482E+01	4.985E+01	3.506E+01	4.322E+01	4.742E+01	5.910E+01	4.551E+01	2.858E+01
F22	Mean	3.621E+03	5.436E+03	3.783E+03	7.031E+03	1.018E+04	1.062E+04	6.129E+03	5.867E+03
	Std	1.339E+03	2.474E+03	2.012E+03	2.192E+03	4.880E+02	1.879E+03	6.638E+02	1.143E+03
F23	Mean	3.137E+03	2.996E+03	2.921E+03	3.160E+03	3.481E+03	3.627E+03	3.087E+03	3.221E+03
	Std	6.828E+01	9.222E+01	9.569E+01	7.780E+01	1.298E+02	9.694E+02	5.576E+01	9.473E+01
F24	Mean	3.324E+03	3.138E+03	3.208E+03	3.271E+03	3.708E+03	3.221E+03	3.326E+03	3.509E+03
	Std	7.902E+01	8.729E+01	1.687E+02	1.047E+02	1.371E+02	7.104E+01	3.554E+01	8.630E+01
F25	Mean	2.964E+03	2.934E+03	2.901E+03	3.641E+03	1.215E+04	3.133E+03	4.902E+03	3.036E+03
	Std	1.534E+01	3.222E+01	2.171E+01	1.908E+02	1.706E+03	6.108E+01	3.824E+02	2.613E+01
F26	Mean	7.187E+03	7.609E+03	5.335E+03	8.275E+03	1.250E+04	7.552E+03	8.093E+03	7.906E+03
	Std	1.248E+03	7.451E+02	1.007E+03	7.075E+02	1.333E+03	2.527E+03	6.319E+02	8.440E+02
F27	Mean	3.695E+03	3.331E+03	3.382E+03	3.619E+03	4.561E+03	3.495E+03	3.495E+03	3.997E+03
	Std	1.239E+02	6.978E+01	8.250E+01	1.666E+02	2.959E+02	2.045E+02	5.570E+01	1.954E+02
F28	Mean	3.352E+03	3.291E+03	3.277E+03	4.798E+03	9.912E+03	3.475E+03	5.282E+03	3.678E+03
	Std	1.990E+01	3.958E+01	6.558E+01	4.073E+02	1.205E+03	6.265E+01	4.098E+02	1.243E+02
F29	Mean	4.525E+03	4.608E+03	4.150E+03	5.347E+03	8.072E+03	5.615E+03	5.099E+03	5.023E+03
	Std	2.131E+02	3.762E+02	2.849E+02	5.052E+02	1.502E+03	7.062E+02	2.183E+02	4.041E+02
F30	Mean	1.448E+06	5.719E+06	1.787E+06	1.672E+08	1.715E+09	5.779E+07	1.964E+08	1.941E+06
	Std	7.583E+05	4.202E+06	7.746E+06	1.327E+08	6.977E+08	4.430E+07	7.200E+07	1.413E+06

Table 16
Results of accuracy test for the composition benchmark functions in CEC2017 with 50 dimensions.

Fun.	Index	CAOA	WOA	PSO	SCA	SHO	SA	ABC	HOA
F21	Mean	2.650E+03	2.825E+03	2.610E+03	3.021E+03	3.346E+03	3.141E+03	3.011E+03	2.731E+03
	Std	5.247E+01	9.705E+01	4.439E+01	6.186E+01	7.156E+01	2.011E+02	4.456E+01	6.414E+01
F22	Mean	9.936E+03	1.106E+04	1.073E+04	1.536E+04	1.726E+04	1.805E+04	1.337E+04	1.035E+04
	Std	9.397E+02	1.155E+03	2.023E+03	8.391E+02	6.051E+02	9.741E+02	5.921E+02	8.175E+02
F23	Mean	3.756E+03	3.515E+03	3.461E+03	3.776E+03	4.448E+03	4.107E+03	3.858E+03	3.928E+03
	Std	1.437E+02	1.898E+02	1.852E+02	1.425E+02	2.065E+02	9.547E+02	9.205E+01	1.584E+02
F24	Mean	4.009E+03	3.621E+03	3.961E+03	3.911E+03	4.949E+03	3.760E+03	4.213E+03	4.268E+03
	Std	1.133E+02	1.355E+02	3.236E+02	1.916E+02	2.076E+02	1.297E+02	1.088E+02	1.593E+02
F25	Mean	3.846E+03	3.147E+03	3.077E+03	8.529E+03	4.244E+04	3.778E+03	1.970E+04	4.956E+03
	Std	8.394E+01	3.979E+01	4.087E+01	9.277E+02	4.575E+03	1.734E+02	2.189E+03	2.206E+02
F26	Mean	1.200E+04	1.234E+04	6.606E+03	1.475E+04	2.303E+04	1.407E+04	1.640E+04	1.183E+04
	Std	7.447E+02	1.273E+03	2.582E+03	6.760E+02	2.149E+03	2.886E+03	1.167E+03	1.093E+03
F27	Mean	5.196E+03	4.184E+03	4.173E+03	5.586E+03	7.244E+03	4.262E+03	4.983E+03	5.753E+03
	Std	3.086E+02	3.662E+02	3.922E+02	6.508E+02	4.715E+02	4.931E+02	1.713E+02	4.313E+02
F28	Mean	4.690E+03	3.497E+03	3.484E+03	8.363E+03	1.803E+04	4.167E+03	1.170E+04	5.815E+03
	Std	1.071E+02	7.724E+01	2.389E+02	7.871E+02	2.002E+03	2.230E+02	7.795E+02	2.361E+02
F29	Mean	6.578E+03	7.039E+03	5.522E+03	1.108E+04	1.221E+05	7.844E+03	1.160E+04	8.484E+03
	Std	5.675E+02	9.453E+02	7.982E+02	2.258E+03	1.267E+05	7.925E+02	1.973E+03	8.217E+02
F30	Mean	9.421E+07	6.748E+07	3.575E+06	1.015E+09	9.478E+09	3.975E+08	3.043E+09	1.496E+08
	Std	1.714E+07	2.286E+07	3.800E+06	5.606E+08	3.013E+09	1.339E+08	6.301E+08	4.347E+07

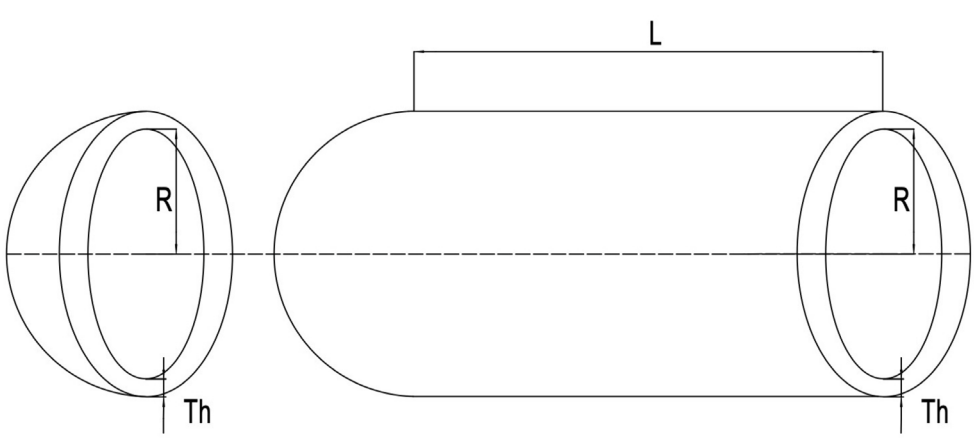

Fig. 4. Schematic of pressure vessel design.

Table 17
Results of accuracy test for the composition benchmark functions in CEC2017 with 100 dimensions.

Fun.	Index	CAOA	WOA	PSO	SCA	SHO	SA	ABC	HOA
F21	Mean	3.760E+03	3.797E+03	3.418E+03	4.261E+03	5.083E+03	4.574E+03	4.481E+03	3.859E+03
	Std	1.253E+02	2.029E+02	1.442E+02	1.719E+02	1.373E+02	1.736E+02	1.018E+02	1.207E+02
F22	Mean	1.989E+04	2.480E+04	2.180E+04	3.301E+04	3.539E+04	3.656E+04	2.912E+04	2.010E+04
	Std	1.139E+03	2.292E+03	2.619E+03	1.422E+03	9.815E+02	1.244E+03	6.635E+02	1.207E+03
F23	Mean	5.332E+03	4.649E+03	5.200E+03	5.287E+03	6.884E+03	7.046E+03	5.811E+03	5.713E+03
	Std	2.960E+02	2.181E+02	5.417E+02	2.334E+02	3.704E+02	2.136E+03	1.939E+02	2.264E+02
F24	Mean	7.906E+03	5.912E+03	8.619E+03	7.511E+03	1.124E+04	5.745E+03	8.977E+03	9.020E+03
	Std	3.319E+02	3.423E+02	1.007E+03	6.006E+02	7.398E+02	2.875E+02	4.436E+02	4.835E+02
F25	Mean	6.713E+03	3.959E+03	3.685E+03	1.823E+04	1.009E+05	8.531E+03	6.580E+04	9.600E+03
	Std	1.612E+02	1.091E+02	1.473E+02	1.526E+03	1.007E+04	9.483E+02	5.334E+03	3.922E+02
F26	Mean	3.024E+04	3.028E+04	2.112E+04	4.088E+04	6.687E+04	3.165E+04	5.286E+04	3.176E+04
	Std	1.832E+03	3.741E+03	7.415E+03	2.435E+03	6.190E+03	3.279E+03	3.251E+03	1.394E+03
F27	Mean	8.871E+03	4.906E+03	5.399E+03	8.488E+03	1.370E+04	4.970E+03	9.659E+03	9.592E+03
	Std	5.842E+02	6.417E+02	8.395E+02	1.258E+03	1.292E+03	5.510E+02	6.616E+02	9.950E+02
F28	Mean	1.119E+04	4.284E+03	4.113E+03	1.805E+04	5.894E+04	9.798E+03	4.321E+04	1.551E+04
	Std	3.380E+02	2.139E+02	6.343E+02	1.482E+03	4.859E+03	2.079E+03	2.779E+03	7.775E+02
F29	Mean	1.169E+04	1.394E+04	9.042E+03	3.127E+04	2.497E+06	1.503E+04	2.912E+05	1.622E+04
	Std	9.038E+02	2.091E+03	8.319E+02	1.360E+04	1.510E+06	1.524E+03	1.319E+05	1.787E+03
F30	Mean	1.551E+09	2.371E+08	2.232E+08	1.139E+10	5.563E+10	1.490E+09	2.974E+10	5.648E+09
	Std	3.923E+08	1.152E+08	3.928E+08	3.870E+09	8.229E+09	3.949E+08	3.667E+09	1.203E+09

Table 18
Results of the Wilcoxon rank-sum test for the composition benchmark functions in CEC2017 with 30 dimensions.

Fun.	Index	CAOA	WOA	PSO	SCA	SHO	SA	ABC	HOA
F21	Mean	2	4	1	6	8	7	5	3
	Std	1	7	3	4	6	8	5	2
F22	Mean	1	3	2	6	7	8	5	4
	Std	4	8	6	7	1	5	2	3
F23	Mean	4	2	1	5	7	8	3	6
	Std	1	4	6	3	7	8	2	5
F24	Mean	5	1	2	4	8	3	6	7
	Std	3	5	8	6	7	2	1	4
F25	Mean	3	2	1	6	8	5	7	4
	Std	1	4	2	6	8	5	7	3
F26	Mean	2	4	1	7	8	3	6	5
	Std	6	3	5	2	7	8	1	4
F27	Mean	6	1	2	5	8	4	3	7
	Std	4	2	3	5	8	7	1	6
F28	Mean	3	2	1	6	8	4	7	5
	Std	1	2	4	6	8	3	7	5
F29	Mean	2	3	1	6	8	7	5	4
	Std	1	4	3	6	8	7	2	5
F30	Mean	1	4	2	6	8	5	7	3
	Std	1	3	4	7	8	5	6	2
Total	Mean	29	26	14	57	78	54	54	48
	Std	23	42	44	52	68	58	34	39

Table 22 presents a comparative analysis of different metaheuristic algorithms applied to the pressure vessel design problem, including CAOA, SHO, SCA, and SA. The results highlight significant differences in the optimal values of key design variables (T_s, T_h, R, and L) and the corresponding optimal vertical deflection. The CAOA algorithm achieved the smallest vertical deflection value (6.110×10^3), indicating superior structural stability while maintaining relatively lower values for T_s and T_h. This suggests that CAOA provides a more material-efficient design without compromising structural performance. In contrast, the SHO algorithm yielded the highest values for T_s and T_h (1.260×10^0 and 6.440×10^0, respectively), which could imply increased material usage. However, despite the thicker material, SHO resulted in the largest vertical deflection (5.134×10^4), suggesting less efficient structural optimization. The SCA algorithm produced the largest values for R and L (4.033×10^1 and 1.999×10^2), indicating a design focused on maximizing volume, though its vertical deflection (1.125×10^4) was higher than that of CAOA. The SA algorithm demonstrated moderate performance across all variables, with neither particularly high nor low values, suggesting a balanced approach between material usage and structural stability. Overall, CAOA appears to offer the most effective trade-off between material efficiency and structural performance, making it the most promising algorithm for the pressure vessel design problem.

4.6.2. *Tension/compression spring design problem*

The tension/compression spring design problem is a widely used benchmark in engineering optimization, where the goal is to minimize the fabrication cost, typically represented by the weight of the spring. The problem involves three decision variables: wire diameter, mean coil diameter, and number of active coils. These parameters must comply with four nonlinear constraints that account for the shear stress limit, surge frequency, maximum deflection, and geometric feasibility of the

Table 19
Results of the Wilcoxon rank-sum test for the composition benchmark functions in CEC2017 with 50 dimensions.

Fun.	Index	CAOA	WOA	PSO	SCA	SHO	SA	ABC	HOA
F21	Mean	2	4	1	6	8	7	5	3
	Std	3	7	1	4	6	8	2	5
F22	Mean	1	4	3	6	7	8	5	2
	Std	5	7	8	4	2	6	1	3
F23	Mean	3	2	1	4	8	7	5	6
	Std	3	6	5	2	7	8	1	4
F24	Mean	5	1	4	3	8	2	6	7
	Std	2	4	8	6	7	3	1	5
F25	Mean	4	2	1	6	8	3	7	5
	Std	3	1	2	6	8	4	7	5
F26	Mean	3	4	1	6	8	5	7	2
	Std	2	5	7	1	6	8	4	3
F27	Mean	5	2	1	6	8	3	4	7
	Std	2	3	4	8	6	7	1	5
F28	Mean	4	2	1	6	8	3	7	5
	Std	2	1	6	7	3	4	8	5
F29	Mean	2	3	1	6	8	4	7	5
	Std	1	5	3	7	8	2	6	4
F30	Mean	3	2	1	6	8	5	7	4
	Std	2	3	1	6	8	5	7	4
Total	Mean	32	26	15	55	79	47	60	46
	Std	25	42	45	51	61	55	38	43

Table 20
Results of the Wilcoxon rank-sum test for the composition benchmark functions in CEC2017 with 100 dimensions.

Fun.	Index	CAOA	WOA	PSO	SCA	SHO	SA	ABC	HOA
F21	Mean	2	3	1	5	8	6	7	4
	Std	3	8	5	6	4	7	1	2
F22	Mean	1	4	3	6	7	8	5	2
	Std	3	7	8	6	2	5	1	4
F23	Mean	4	1	2	3	7	8	6	5
	Std	5	2	7	4	6	8	1	3
F24	Mean	4	2	5	3	8	1	6	7
	Std	2	3	8	6	7	1	4	5
F25	Mean	3	2	1	6	8	4	7	5
	Std	3	1	2	6	8	5	7	4
F26	Mean	2	3	1	6	8	4	7	5
	Std	2	6	8	3	7	5	4	1
F27	Mean	5	1	3	4	8	2	7	6
	Std	2	3	5	7	8	1	4	6
F28	Mean	4	2	1	6	8	3	7	5
	Std	2	1	3	5	8	6	7	4
F29	Mean	2	3	1	6	8	4	7	5
	Std	2	5	1	6	8	3	7	4
F30	Mean	4	2	1	6	8	3	7	5
	Std	2	1	3	7	8	4	6	5
Total	Mean	31	23	19	51	78	43	66	49
	Std	26	37	50	56	66	45	42	38

spring design. The problem reflects real-world trade-offs between performance and material usage in mechanical systems. To ensure feasible solutions, the penalty method is employed to incorporate constraint violations directly into the objective function. The simulation process follows the same settings as the previous problem, using a population size of 30, 500 iterations, and averaging over 30 independent runs to reduce randomness and improve statistical significance. The structural diagram of the spring is illustrated in Fig. 5.

$$\min_{d,D,N} \quad (N+2)Dd^2 \tag{20}$$

Subject to:

$$1 - \frac{D^3 N}{71785 d^4} \le 0 \tag{21}$$

$$\frac{4D^2 - dD}{12566(Dd^3 - d^4)} + \frac{1}{5108d^2} \le 0 \tag{22}$$

$$1 - \frac{140.45d}{D^2 N} \le 0 \tag{23}$$

$$\frac{d+D}{1.5} - 1 \le 0 \tag{24}$$

$$0.05 \le d \le 2.0; 0.15 \le D \le 1.3 \tag{25}$$

$$2.0 \le N \le 15.0 \tag{26}$$

Table 23 presents a comparative analysis of various metaheuristic algorithms applied to the tension/compression spring design problem,

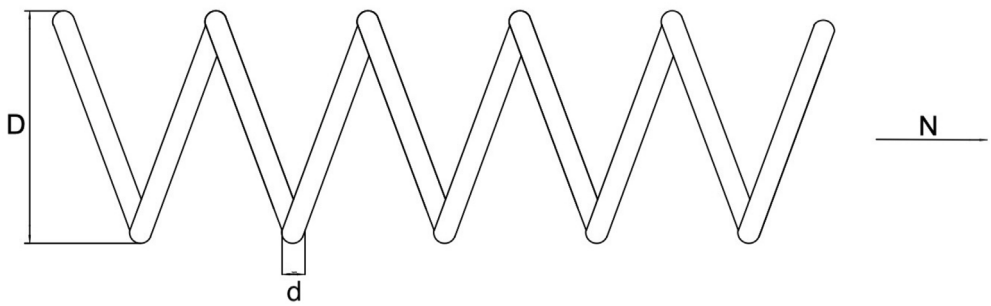

Fig. 5. Schematic of tension/compression spring design.

Table 21
Average Runtime (in seconds) for Each Algorithm Across Dimensions.

Algorithm	Dimension	AvgTime (s)
CAOA	10	0.0044
AVOA	10	0.0058
HLOA	10	0.0247
GTO	10	0.0073
MGO	10	0.0199
CAOA	30	0.0037
AVOA	30	0.0130
HLOA	30	0.0737
GTO	30	0.0250
MGO	30	0.0666
CAOA	50	0.0067
AVOA	50	0.0220
HLOA	50	0.1245
GTO	50	0.0573
MGO	50	0.1721

Table 22
Comparison results for pressure vessel design problem.

Metah.	Optimal values of variables				Opt. Vert. defl.
	T_s	T_h	R	L	
CAOA	8.819E−01	4.391E−01	4.565E+01	1.373E+02	6.110E+03
SHO	1.260E+00	6.440E+00	6.523E+01	1.000E+01	5.134E+04
SCA	7.790E−01	2.238E+00	4.033E+01	1.999E+02	1.125E+04
SA	1.129E+00	1.557E+00	5.451E+01	9.038E+01	1.343E+04

Table 23
Comparison results for tension/compression spring design problem.

Metah.	Optimal values of variables			Opt. Vert. defl.
	d	D	N	
CAOA	5.260E−02	3.778E−01	1.030E+01	1.290E−02
SHO	5.940E−02	5.716E−01	4.780E+00	1.370E−02
SCA	6.350E−02	7.109E−01	3.257E+00	1.510E−02
SA	1.101E−01	1.195E+00	6.299E+00	1.203E−01

including CAOA, SHO, SCA, and SA. The results reveal distinct differences in the optimal values of the design variables (d, D, and N) and the corresponding optimal vertical deflection. The CAOA algorithm achieved the smallest vertical deflection (1.290×10^{-2}), indicating superior structural performance while maintaining relatively low values for d, D, and N. This suggests that CAOA offers an efficient design with minimal material usage while ensuring structural integrity. In contrast, the SA algorithm resulted in the highest values for all variables ($d = 1.101 \times 10^{-1}$, $D = 1.195 \times 10^{0}$, and $N = 6.299 \times 10^{0}$), along with the largest vertical deflection (1.203×10^{-1}), indicating an inefficient design with excessive material usage and suboptimal deformation control. The SCA algorithm produced moderate values for d, D, and N, but with a higher vertical deflection (1.510×10^{-2}) than both CAOA and SHO, suggesting less effective optimization. SHO demonstrated better performance compared to SCA and SA, achieving a relatively low deflection (1.370×10^{-2}) with moderate design variable values. Overall, CAOA outperformed the other algorithms in terms of minimizing vertical deflection while maintaining material efficiency, making it the most promising algorithm for the tension/compression spring design problem.

5. Discussion

The experimental results have extensively proved CAOA superiority, exhibiting its robustness, accuracy, and scalability over varying optimization scenarios. In unimodal and simple multimodal functions, CAOA consistently outperformed its peers concerning both mean error and stability. Results in 30, 50, and 100 dimensions have provided strong evidence that CAOA managed a good convergence to the global optimum, as shown by its significantly lower mean errors compared to WOA, PSO, and several other contenders. In particular, CAOA's substantial error reduction on both F1 and F5 proves its powerful exploitation ability in unimodal landscapes while simultaneously exploring multimodal search spaces with ease. The generally low standard deviation values indicate that CAOA maintains a stable search trajectory, confirming reliability across multiple runs. These findings suggest that CAOA adeptly balances exploration and exploitation-an essential trait in optimizing complex search landscapes.

With increasing dimensionality of the problems, CAOA's convergence behavior continues to display great prowess and outstand the performance of alternative algorithms. Observed performance trends in 50- and 100-dimensional cases suggest robust scaling by CAOA, handling high-dimensional search spaces with only minor losses in accuracy. It is noteworthy that CAOA yielded lower mean errors and standard deviations in benchmark functions like F6 and F8, cementing their operational capability to maintain stalemate performance while optimizing increasingly complex problems. These observations were supported by the Wilcoxon rank-sum tests, with CAOA acquiring the least mean rank in varying dimensional settings. The statistical significance of these results, with p-values below 0.05, confirms that CAOA's performance improvements are not incidental but rather a result of its effective optimization strategies.

With regard to the hybrid benchmark functions integrating multiple complex landscapes, CAOA is persistently superior in its convergence accuracy and robustness. For each dimension, the algorithm produced the smallest mean error values over a majority of the test functions, outperforming WOA, PSO, and other competitors. The improvement in the accuracy of functions such as F11, F12, and F15 shows CAOA's versatility in the presence of highly intricate search spaces. In addition, CAOA yields a lower standard deviation than its rival peers, further fortifying its stability in respect to complex optimization problems. Upon increasing dimensions, CAOA maintains optimization efficiency, attaining lower mean errors than competitors' approaches for hybrid functions. The statistical analysis agrees with this conclusion whereby CAOA stayed with lower mean ranks against other algorithms, and its significant advantage has been reflected on the Wilcoxon rank-sum test.

These results imply that CAOA addresses the difficulties presented by hybrid landscapes in a superior manner, where it gains ground during the paper in terms of both accuracy and search stability.

CAOA maintains its edge over the other methods when attention is paid to composition functions, which add yet another level of complexity due to the interaction of multiple landscapes and their intricacies. The experimental results validate that CAOA had an edge over the competing algorithms in low to high-dimensional contexts, always attaining significantly reduced mean errors and showing a degree of convergence that was tougher than that of the competing methods. The clear dominance of CAOA along the lines of F21 and F24, extremely difficult functions, demonstrates that, on one hand, CAOA passes through rugged fitness landscapes with high efficiency in refining solutions. With rising dimensionality, CAOA ultimately secures the least mean error and standard deviations, hence marking its robustness in tackling high-dimensional composition functions. Statistical meaning analyses provide further credibility for these findings since CAOA was consistently the best across each dimension with respect to the total mean ranks, proving its dominance in the composite search spaces.

Substance-wise, CAOA shines by generating intriguing results on real-life engineering optimization problems, evidenced in pressure vessel designs and tension/compression spring design problems. Findings indicate that CAOA had structural stability, with least material wastage, where SA and SHO and SCA fall short when it comes to the optimization of structures. In the pressure vessel design problem, CAOA managed to get the minimal deflection while also employing less material: a classic case of happily efficient designs. In designing tension and compression springs, CAOA continues administering the minimal vertical deflection further factored for optimizing mechanical structures with ease. Such a conclusion hints that, while CAOA succeeds on a brainstorm of theoretical benchmark functions, it also has much-prized practicality when enlisted for applications in engineering design.

The experimental results substantiate how CAOA outperformed state-of-the-art metaheuristic algorithms in all optimization landscapes. Its increasing accuracy, stability, and scalability lend themselves to CAOA being a highly vehicular optimization method applied in benchmark functions and real-world applications. The Wilcoxon rank-sum tests reaffirm CAOA's effectiveness with solid statistical verification that the improvement is substantial rather than coincidental. The combined results have established CAOA as a robust optimization algorithm equivalent to solving several difficult optimization problems.

6. Conclusion and future work

This paper introduced the Crocodile Ambush Optimization Algorithm (CAOA), a novel metaheuristic inspired by the hunting strategies of crocodiles. The algorithm incorporates several adaptive mechanisms, including energy decay modeling, stochastic leader selection, and threshold-triggered reinitialization, to balance global exploration and local exploitation.

Extensive experiments on thirteen benchmark functions from the CEC 2017 test suite demonstrated that CAOA achieves superior or highly competitive results compared to recent state-of-the-art algorithms such as AVOA, HLOA, GTO, MGO, FFA, and POA. Time complexity evaluations confirmed the algorithm's computational efficiency, particularly in lower dimensions. Furthermore, two constrained real-world engineering problems were used to validate CAOA's effectiveness in practical scenarios, showing that it can generate feasible, high-quality solutions under strict constraints.

The key contributions of this paper are threefold: (1) the design of a new crocodile-inspired optimization algorithm with multiple adaptive behaviors; (2) an in-depth empirical evaluation across benchmark and real-world problems; and (3) comparative analyses against multiple newly proposed metaheuristics in terms of performance and runtime.

For future work, I plan to extend CAOA to multi-objective optimization problems, investigate its performance in dynamic and large-scale environments, and explore hybridization strategies that integrate local search or domain-specific heuristics to further enhance convergence and precision.

Declaration on AI Tools

This manuscript was written entirely by the author without the use of generative AI technologies.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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